Chapter 09

Radiative Cooling, Shielding, and Multiphysics

The Mixed-Mode Frontier: Where Conduction Meets Radiation

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

LO-09.1 Derive the net radiative cooling power balance including atmospheric absorption, solar heating, and convective parasitic losses, and identify the spectral design requirements for sub-ambient cooling.

LO-09.2 Evaluate and compare material systems for radiative cooling—polymer films, photonic crystals, and chalcogenide metasurfaces—using quantitative figures of merit including P_{cool}, spectral selectivity, durability, and manufacturing cost.

LO-09.3 Design radiation shields with specified attenuation factors using multilayer and metamaterial approaches, and integrate them into the Mode-Path Matrix as modified G_{rad} elements via Recipe 09.R2.

LO-09.4 Formulate the multiphysics cloaking problem as a dual-mode challenge requiring simultaneous conduction cloaking (Part II tools) and radiation shielding (Part III tools), with the instrument-consistent metric $R_{\mathrm{cloak,rad}}$ (Metric 09.M).

LO-09.5 Explain why linear superposition $G_{\mathrm{total}} = G_{\mathrm{cond}} + G_{\mathrm{rad}}$ fails in the mixed-mode regime, and derive the self-consistency requirement that motivates iterative coupling.

LO-09.6 Apply Algorithm 09.A (the coupled conduction–radiation solver) to mixed-mode paths where $0.1 < \Gamma < 10$, including dual convergence criteria, error estimation, and fallback to Newton–Raphson iteration.

LO-09.7 Construct the coupled thermal network for a semiconductor fabrication system, solve it self-consistently, and demonstrate iterative convergence via the Mode-Path Matrix.

LO-09.8 Identify semiconductor fabrication processes that operate in the mixed-mode regime and select the appropriate coupled design tools using the Tool Translation Table (Chapter 4).

Abbreviations and Nomenclature

Abbreviation	**Meaning**
LWIR	Long-wave infrared (8–14 μm)
MWIR	Mid-wave infrared (3–5 μm)
MPM	Mode-Path Matrix
EMT	Effective medium theory
FEA	Finite element analysis
FOM	Figure of merit
PECVD	Plasma-enhanced chemical vapor deposition
RTP	Rapid thermal processing
CVD	Chemical vapor deposition
PDMS	Polydimethylsiloxane
PMMA	Polymethyl methacrylate
AR	Anti-reflection
PGM	Precision glass molding
SPhP	Surface phonon polariton
TPV	Thermophotovoltaic
ESC	Electrostatic chuck

Symbol	**Meaning**	**SI Unit**
P_{cool}	Net radiative cooling power	$\mathrm{W\,m^{-2}}$
P_{rad}	Emitted radiative power	$\mathrm{W\,m^{-2}}$
P_{atm}	Absorbed atmospheric radiation	$\mathrm{W\,m^{-2}}$
P_{sol}	Absorbed solar power	$\mathrm{W\,m^{-2}}$
P_{conv}	Convective parasitic heat gain	$\mathrm{W\,m^{-2}}$
$\varepsilon(\lambda)$	Spectral emissivity	—
ε_{eff}	Effective (band-averaged) emissivity	—
α_{sol}	Solar absorptance	—
$B(\lambda, T)$	Planck spectral radiance	$\mathrm{W\,m^{-2}\,sr^{-1}\,m^{-1}}$
$\tau_{\text{atm}}(\lambda)$	Atmospheric spectral transmittance	—
Γ	Mode dominance ratio ($G_{\text{rad}}/G_{\text{cond}}$)	—
G_{cond}	Conduction conductance	$\mathrm{W\,K^{-1}}$
G_{rad}	Radiation conductance	$\mathrm{W\,K^{-1}}$
G_{total}	Total conductance	$\mathrm{W\,K^{-1}}$
G_{conv}	Convective conductance	$\mathrm{W\,K^{-1}}$
$G_{\text{rad}}^{\text{shielded}}$	Shielded radiation conductance	$\mathrm{W\,K^{-1}}$
$\boldsymbol{\kappa}$	Thermal conductivity tensor	$\mathrm{W\,m^{-1}\,K^{-1}}$
h_{conv}	Convective heat transfer coefficient	$\mathrm{W\,m^{-2}\,K^{-1}}$
h_{crit}	Critical convective coefficient for sub-ambient cooling	$\mathrm{W\,m^{-2}\,K^{-1}}$
R_{cloak}	Cloaking efficiency metric	—
$R_{\text{cloak,cond}}$	Conduction cloaking efficiency	—
$R_{\text{cloak,rad}}$	Radiation cloaking efficiency	—
R_{multi}	Multiphysics cloaking efficiency	—
F_{12}	View factor	—
n	Refractive index	—
σ_{SB}	Stefan–Boltzmann constant	$\mathrm{W\,m^{-2}\,K^{-4}}$
ω	Under-relaxation parameter	—
ϵ_T	Temperature convergence tolerance	K
ϵ_E	Energy residual convergence tolerance	W

Subscript/Superscript	Convention
atm	Atmospheric (8–13 μm transparency window)
sol	Solar spectrum
cool	Net cooling
rad	Radiative
cond	Conductive
conv	Convective
eff	Effective (band-averaged or system-level)
surf	Surface
sky	Effective sky (cosmic background)
multi	Multiphysics (simultaneous cond + rad)
crit	Critical (threshold value)

Numbering convention for this chapter: Design rules use the **DR-09.n** prefix. Topological design rules inherited from Chapter 4 retain the **DR-T.n** prefix. Worked examples use **WE-09.n**. Learning objectives use **LO-09.n**. MPM integration recipes use **Recipe 09.Rn**.

Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

A complete mixed-mode design methodology—from radiative cooling physics through multiphysics cloaking to coupled conduction–radiation network analysis—that handles the $0.1 < \Gamma < 10$ regime where neither Part II nor Part III tools alone suffice, with a formal iterative solver (Algorithm 09.A), dual convergence criteria, error estimation, and worked examples demonstrating the full Mode-Path Matrix workflow.

To help you navigate, Figure 09.1 shows the chapter "spine" (workflow) and how each section feeds into the final mixed-mode design toolkit.

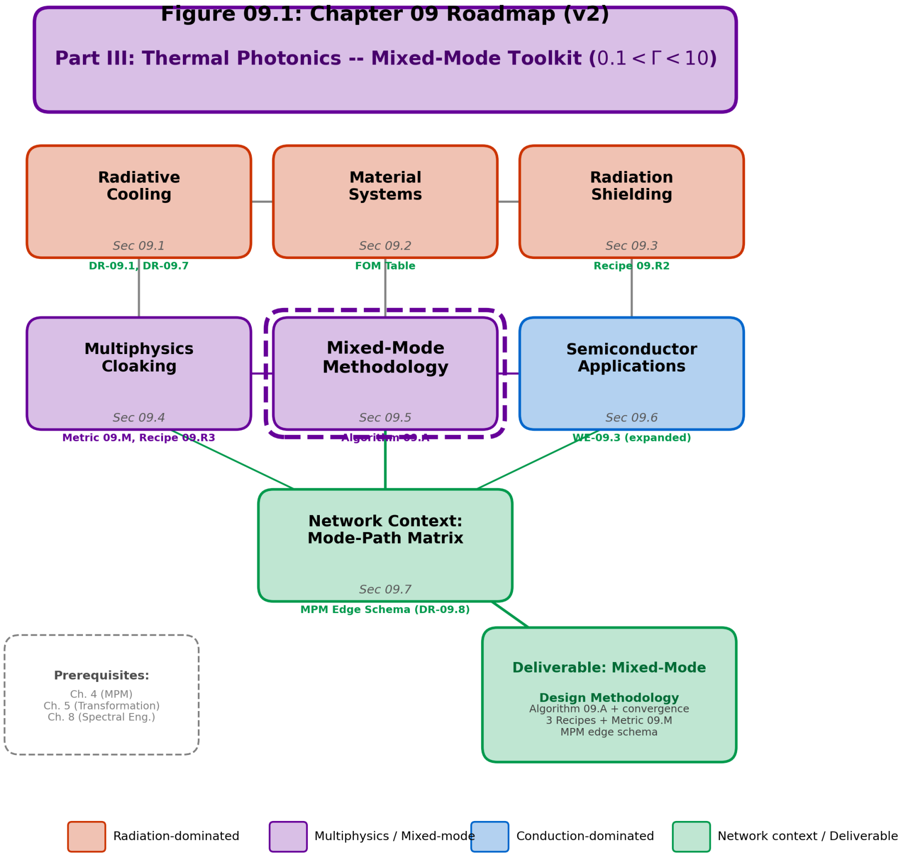


Figure 09.1: Chapter 09 roadmap (v2). The narrative spine begins with daytime radiative cooling physics (§09.1), establishes material systems for spectral engineering (§09.2), then develops radiation shielding (§09.3) and multiphysics cloaking (§09.4) as applications requiring dual-mode control. Section 09.5 derives the self-consistency requirement and formalizes Algorithm 09.A for $0.1 < \Gamma < 10$. Section 09.6 presents the expanded PECVD semiconductor case study, and Section 09.7 integrates the coupled analysis into the Mode-Path Matrix framework via the standardized edge schema. Purple boxes indicate multiphysics/mixed-mode content; red boxes indicate radiation-dominated content; blue boxes indicate conduction-dominated content. The green network context box connects to the architectural framework of Chapter 4. Three MPM integration recipe boxes (R1–R3) and one formal algorithm box (09.A) are new in v2.

Why this chapter is essential. Parts II and III of this book provide complementary toolkits: Diffusionics for conduction-dominated paths ($\Gamma < 0.1$) and Thermal Photonics for radiation-dominated paths ($\Gamma > 10$). These toolkits are powerful when the Mode-Path Matrix clearly assigns a path to one regime. But what happens when $0.1 < \Gamma < 10$?

In the mixed-mode band, conduction and radiation carry comparable fractions of the total heat flux. Modifying one conductance—say, adding a selective emitter (Part III)—changes the temperature distribution, which in turn changes the conduction conductance (via temperature-dependent κ) and the radiation conductance (via the T^3 dependence of G_{rad}). The two modes are *coupled*, and the coupled system must be solved *self-consistently*. A naive superposition $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}}(T_0)$ evaluated at the initial temperature T_0 fails because G_{rad} depends on the very temperature distribution one is trying to compute (§09.5 provides the formal derivation).

Central methodology preview. The core algorithmic contribution of this chapter is **Algorithm 09.A** (§09.5.3): a coupled conduction–radiation solver that initializes with a conduction-only solution, computes $G_{\mathrm{rad}}(T)$, re-solves the network with under-relaxation, and iterates until *dual* convergence criteria are satisfied (temperature residual $< \epsilon_T$ *and* energy residual $< \epsilon_E$). This algorithm applies to every mixed-mode system in the book and produces, as its output, an updated Mode-Path Matrix with self-consistent Γ values for all paths.

This chapter addresses three questions that earlier chapters cannot:

1. How do we engineer radiative cooling systems that exploit the atmospheric transparency window? (Application of Ch. 8 spectral tools to a specific, high-impact problem.)
2. How do we cloak or shield a thermal signature when both conduction and radiation pathways exist? (The multiphysics cloaking problem.)
3. How do we analyze and design systems in the mixed-mode regime where the Part II and Part III toolkits must be combined? (The central methodological contribution.)

Pain Point

Semiconductor fabrication processes in the 300–600 °C range—PECVD, thermal oxidation, medium-temperature CVD—operate squarely in the mixed-mode regime ($0.1 < \Gamma < 10$). An engineer who models these systems using conduction-only tools (ignoring radiation) or radiation-only tools (ignoring conduction) introduces systematic errors of 10–90% in thermal resistance predictions (DR-09.6, Table 09.8). These errors propagate directly into temperature non-uniformity specifications, causing yield loss that could have been prevented by proper mixed-mode analysis. This chapter provides the missing methodology.

Reader Recipe

MPM-first workflow for this chapter:

1. Compute Γ for the path of interest using Chapter 4 methods. If $\Gamma < 0.1$ or $\Gamma > 10$, use Part II or Part III tools directly. This chapter applies when $0.1 < \Gamma < 10$.
2. For radiative cooling applications (§09.1–§09.2): compute the net cooling power balance, verify the convection hard gate (DR-09.7), select materials from the catalog (Table 09.5), and design the spectral profile using Ch. 8 tools.
3. For shielding/cloaking (§09.3–§09.4): identify both conduction and radiation pathways; design independent solutions for each mode, then verify compatibility using the instrument-consistent metric $R_{\mathrm{cloak,rad}}$ (Metric 09.M).
4. For general mixed-mode analysis (§09.5): construct the coupled network, apply Algorithm 09.A, and verify convergence (DR-09.5).
5. Update the Mode-Path Matrix with the coupled conductances using the MPM edge schema (§09.7).

09.1 Daytime Radiative Cooling Physics

Radiative cooling is the process by which a surface sheds thermal energy as electromagnetic radiation to a cold sink. At night, this mechanism has been exploited for millennia: desert cultures observed that water could freeze in shallow trays despite ambient air temperatures above 0 °C. The cold sink is outer space, at an effective temperature of ∼ 3 K, accessible through a spectral window in the Earth's atmosphere.

The engineering challenge is to achieve radiative cooling *during the day*, when the surface is illuminated by ∼ 1000 $\mathrm{W\,m^{-2}}$ of solar radiation. This requires a surface that simultaneously:

1. *Emits strongly* in the atmospheric transparency window (8–13 μm), maximizing heat rejection to space;
2. *Reflects strongly* across the solar spectrum (0.3–2.5 μm), minimizing solar heat gain;
3. *Emits weakly* outside the atmospheric window, minimizing reabsorption from atmospheric greenhouse gases.

This is a spectral engineering problem of the type addressed in Chapter 8, but with a specific physical context that couples the radiative exchange to convective and conductive parasitic losses. The resulting system operates at $\Gamma \approx 1$—squarely in the mixed-mode band.

09.1.1 The Atmospheric Transparency Window

Earth's atmosphere is largely opaque to thermal infrared radiation due to absorption by H_2O and CO_2. However, between approximately 8 and 13 μm, a transparency window exists where the atmospheric transmittance exceeds 0.8 under clear-sky conditions (Figure 09.2).

Figure 09.2: Atmospheric Transparency Window and Thermal Emission

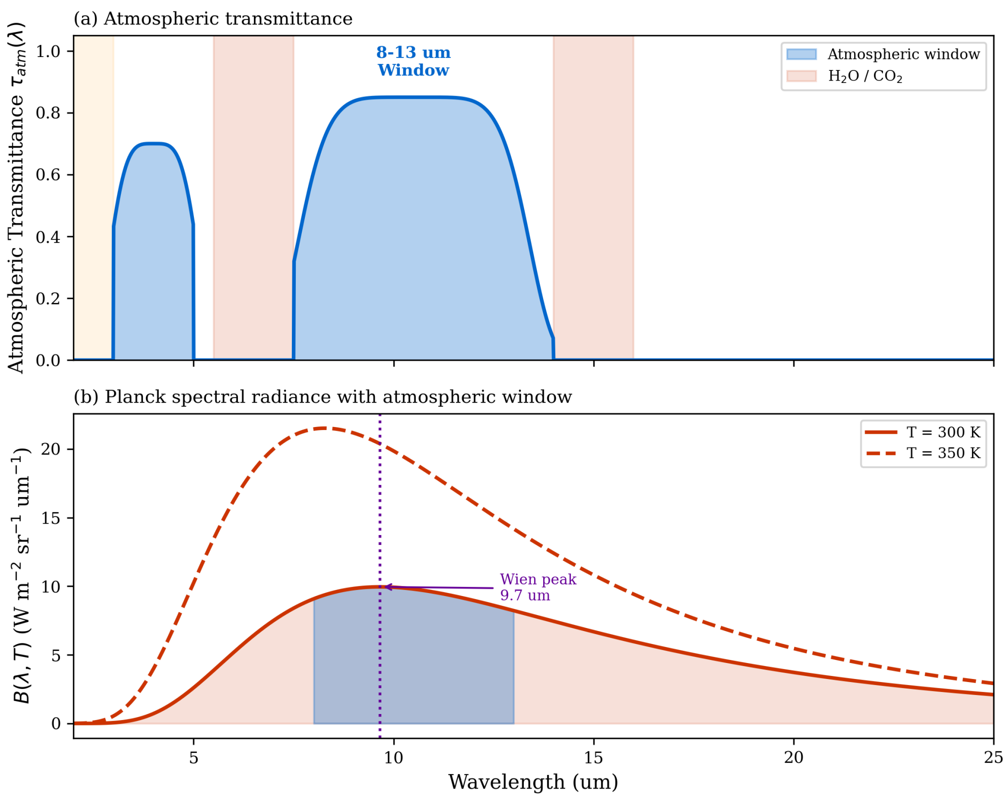


Figure 09.2: Atmospheric transparency and Planck spectra. **(a)** Atmospheric transmittance $\tau_{\mathrm{atm}}(\lambda)$ showing the 8–13 μm transparency window. The dashed curves show Planck spectral radiance $B(\lambda, T)$ at 300 K and 260 K, illustrating that the window captures approximately 28% of the total blackbody emission at ambient temperature. **(b)** The spectral overlap between the Planck distribution and the atmospheric window defines the maximum radiative cooling power. Shaded blue region: power accessible through the window.

At 300 K, the Planck distribution peaks near $\lambda_{\text{peak}} = 2898/300 \approx 9.7\ \mu\text{m}$ (Wien's law), which falls within the atmospheric window. This fortunate coincidence means that a significant fraction of the blackbody emission at ambient temperature can escape to space.

The fraction of blackbody power within the 8–13 μm band is approximately:

$$f_{\text{window}} = \frac{\int_8^{13} B(\lambda, 300\ \text{K})\, \mathrm{d}\lambda}{\int_0^{\infty} B(\lambda, 300\ \text{K})\, \mathrm{d}\lambda} \approx 0.28 \tag{09.1}$$

so that $f_{\text{window}} \times \sigma_{\text{SB}} T^4 = 0.28 \times 459 \approx 130\ \text{W}\,\text{m}^{-2}$ is available for radiative cooling through the window.

09.1.2 Net Cooling Power Balance

The net cooling power per unit area of a radiative cooler at surface temperature T_{surf} under an atmosphere at temperature T_{atm} is:

$$\boxed{P_{\text{cool}}(T_{\text{surf}}) = P_{\text{rad}}(T_{\text{surf}}) - P_{\text{atm}}(T_{\text{atm}}) - P_{\text{sol}} - P_{\text{conv}}(T_{\text{surf}}, T_{\text{atm}})} \tag{09.2}$$

where the four terms are:

Emitted radiative power.

$$P_{\text{rad}}(T_{\text{surf}}) = \int_0^{\infty} \varepsilon(\lambda)\, \pi\, B(\lambda, T_{\text{surf}})\, \mathrm{d}\lambda \tag{09.3}$$

Absorbed atmospheric radiation.

$$P_{\text{atm}}(T_{\text{atm}}) = \int_0^{\infty} \varepsilon(\lambda)\, [1 - \tau_{\text{atm}}(\lambda)]\, \pi\, B(\lambda, T_{\text{atm}})\, \mathrm{d}\lambda \tag{09.4}$$

where $[1 - \tau_{\text{atm}}(\lambda)]$ is the atmospheric emissivity at wavelength λ (Kirchhoff's law applied to the atmosphere as a distributed emitter).

Absorbed solar radiation.

$$P_{\text{sol}} = \alpha_{\text{sol}}\, I_{\text{sol}} \tag{09.5}$$

where $I_{\text{sol}} \approx 1000\ \text{W}\,\text{m}^{-2}$ (AM1.5 global) and α_{sol} is the solar-weighted absorptance of the cooler surface.

Convective parasitic heat gain.

$$P_{\text{conv}} = h_{\text{conv}}\, (T_{\text{atm}} - T_{\text{surf}}) \tag{09.6}$$

where h_{conv} is the convective heat transfer coefficient. Note that $P_{\text{conv}} > 0$ when $T_{\text{surf}} < T_{\text{atm}}$ (the cooler absorbs heat from the warmer air), making convection a parasitic loss that opposes radiative cooling.

Figure 09.3 shows the power balance diagram.

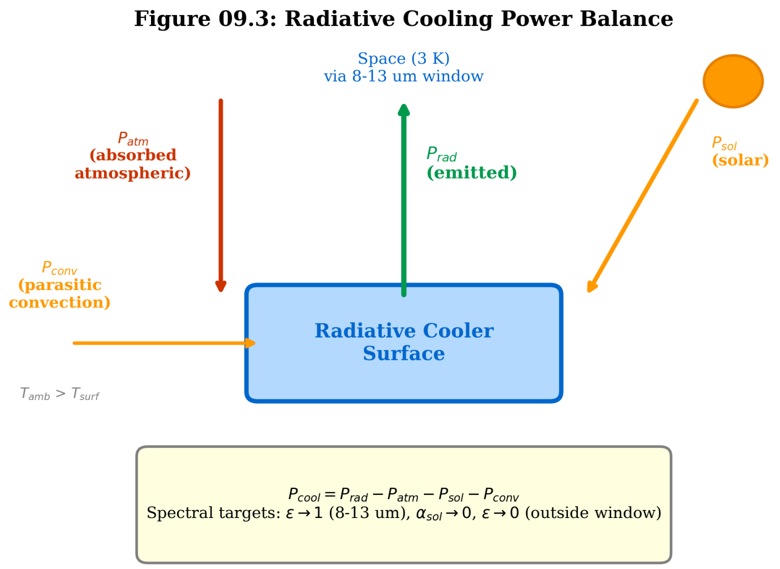


Figure 09.3: Radiative cooling power balance diagram. Arrows indicate heat flux direction. P_{rad} (upward, through atmospheric window) is the useful cooling term. P_{atm} (downward), P_{sol} (downward), and P_{conv} (into surface when $T_{\mathrm{surf}} < T_{\mathrm{atm}}$) are parasitic loads. Net cooling $P_{\mathrm{cool}} = P_{\mathrm{rad}} - P_{\mathrm{atm}} - P_{\mathrm{sol}} - P_{\mathrm{conv}}$.

09.1.3 Spectral Design Requirements

The cooling power balance (Eq. (09.2)) reveals three spectral design knobs:

1. **Maximize $\varepsilon(\lambda)$ in the atmospheric window (8–13 μm)** to maximize P_{rad}. The ideal cooler has $\varepsilon = 1$ in this band.
2. **Minimize α_{sol} across the solar spectrum (0.3–2.5 μm)** to minimize P_{sol}. The ideal cooler has $\alpha_{\mathrm{sol}} = 0$ (perfect solar reflector).
3. **Minimize $\varepsilon(\lambda)$ outside the atmospheric window** to minimize P_{atm}. Outside the window, atmospheric emissivity approaches unity, so any surface emission is fully reabsorbed.

The ideal spectral emissivity profile is therefore a rectangular function: $\varepsilon = 1$ for $8 \le \lambda \le 13$ μm, and $\varepsilon = 0$ elsewhere. Real materials approximate this profile to varying degrees, as discussed in §09.2.

09.1.4 Equilibrium Temperature and the Convection Hard Gate

At thermal equilibrium, $P_{\mathrm{cool}} = 0$. For small departures from ambient ($T_{\mathrm{surf}} = T_{\mathrm{atm}} - \Delta T$), linearizing gives:

$$0 = P_{\mathrm{cool}}(T_{\mathrm{atm}}) - \underbrace{\left(4\varepsilon_{\mathrm{eff}}\sigma_{\mathrm{SB}}T_{\mathrm{atm}}^3 + h_{\mathrm{conv}}\right)}_{g_{\mathrm{total}}} \Delta T \tag{09.7}$$

where g_{total} is the total heat transfer coefficient per unit area (units: $\mathrm{W\,m^{-2}\,K^{-1}}$) combining the linearized radiation and convection contributions. The equilibrium sub-ambient temperature depression is:

$$\Delta T = \frac{P_{\mathrm{cool}}(T_{\mathrm{atm}})}{g_{\mathrm{total}}} = \frac{P_{\mathrm{cool}}(T_{\mathrm{atm}})}{4\varepsilon_{\mathrm{eff}}\sigma_{\mathrm{SB}}T_{\mathrm{atm}}^3 + h_{\mathrm{conv}}} \tag{09.8}$$

Sub-ambient cooling ($\Delta T > 0$) requires $P_{\text{cool}}(T_{\text{atm}}) > 0$, which imposes an upper limit on the convective coefficient. Setting $\Delta T = 0$ gives the critical convective threshold:

$$\boxed{h_{\text{crit}} = \frac{P_{\text{cool}}(T_{\text{atm}})}{\Delta T_{\text{target}}}} \tag{09.9}$$

More precisely, sub-ambient cooling of at least ΔT_{target} requires:

$$h_{\text{conv}} < \frac{P_{\text{cool}}(T_{\text{atm}})}{\Delta T_{\text{target}}} - 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T_{\text{atm}}^3 \tag{09.10}$$

DR-09.7 — Convection Hard Gate for Sub-Ambient Cooling

Before optimizing the spectral emissivity profile, verify that the convective environment permits sub-ambient operation. For a target temperature depression ΔT_{target}:

$$h_{\text{conv}} < h_{\text{crit}} = \frac{P_{\text{cool}}(T_{\text{atm}})}{\Delta T_{\text{target}}} - 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T_{\text{atm}}^3 \tag{DR-09.7}$$

For a typical chalcogenide metasurface cooler ($P_{\text{cool}} \approx 65$ W m^{-2} at 300 K, $\varepsilon_{\text{eff}} = 0.95$) targeting $\Delta T = 5\,^\circ$C: $h_{\text{crit}} = 65/5 - 5.53 \approx 7.5$ W m^{-2} K^{-1}. Typical outdoor $h_{\text{conv}} = 5$–10 W m^{-2} K^{-1} (light to moderate wind). *If $h_{\text{conv}} > h_{\text{crit}}$, no amount of spectral optimization can achieve the target ΔT; the designer must first reduce convective loss* (e.g., IR-transparent polyethylene wind shield with $\tau > 0.9$ in 8–13 μm, or vacuum enclosure).

Radiation Mode (Thermal Photonics)

Mode-Path Matrix context for radiative cooling. A radiative cooler is a three-path system in the MPM: (1) surface $\to$ sky via radiation through the atmospheric window (G_{rad}), (2) surface $\to$ air via convection (G_{conv}), and (3) surface $\to$ sun via solar absorption (P_{sol}, a fixed source term). The mode dominance ratio for the surface-to-sky path is $\Gamma = G_{\text{rad}}/G_{\text{conv}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3/h_{\text{conv}}$. At $T = 300$ K with $\varepsilon_{\text{eff}} = 0.95$ and $h_{\text{conv}} = 5$ W m^{-2} K^{-1}: $\Gamma = 5.53/5 = 1.1$, confirming the mixed-mode character.

09.1.5 Worked Example: WE-09.1—As_2S_3 Radiative Cooler Design

Problem: Design a daytime radiative cooler using As_2S_3 nanodisks on Al substrate. Calculate the net cooling power at $T_{\text{surf}} = 300$ K and the equilibrium temperature depression ΔT below ambient for $h_{\text{conv}} = 5$ W m^{-2} K^{-1}.

Given:

- $\varepsilon_{\text{eff}} = 0.95$ in the 8–13 μm window; $\varepsilon = 0.02$ outside.
- $\alpha_{\text{sol}} = 0.04$.
- $I_{\text{sol}} = 1000$ W m^{-2}.
- $T_{\text{atm}} = 300$ K, clear sky.
- $h_{\text{conv}} = 5$ W m^{-2} K^{-1}.

Step 1: Emitted radiative power. The power emitted through the atmospheric window (approximating the window integral):

$$P_{\text{rad}} = \varepsilon_{\text{eff}} \cdot \pi \int_8^{13} B(\lambda, 300)\, \mathrm{d}\lambda + 0.02 \cdot \pi \int_{\text{outside}} B(\lambda, 300)\, \mathrm{d}\lambda \tag{09.11}$$

$$\approx 0.95 \times 130 + 0.02 \times (459 - 130) = 123.5 + 6.6 = 130.1 \text{ W m}^{-2} \tag{09.12}$$

where $\sigma_{\text{SB}}T^4 = 459 \text{ W m}^{-2}$ is the total blackbody emissive power at 300 K, and the 8–13 μm band contains approximately 130 W m^{-2} (28% of blackbody emission).

Step 2: Absorbed atmospheric radiation. For clear-sky conditions with mean atmospheric transmittance $\bar{\tau}_{\text{atm}} \approx 0.85$ in the window:

$$P_{\text{atm}} = \varepsilon_{\text{eff}} \cdot (1 - \bar{\tau}_{\text{atm}}) \cdot \pi \int_8^{13} B(\lambda, 300)\, \mathrm{d}\lambda + 0.02 \times 1.0 \times 329 \tag{09.13}$$

$$\approx 0.95 \times 0.15 \times 130 + 0.02 \times 329 = 18.5 + 6.6 = 25.1 \text{ W m}^{-2} \tag{09.14}$$

Step 3: Absorbed solar radiation.

$$P_{\text{sol}} = 0.04 \times 1000 = 40 \text{ W m}^{-2} \tag{09.15}$$

Step 4: Net cooling at ambient temperature. At $T_{\text{surf}} = T_{\text{atm}} = 300$ K (no convective exchange):

$$P_{\text{cool}}(300 \text{ K}) = 130.1 - 25.1 - 40 - 0 = 65.0 \text{ W m}^{-2} \tag{09.16}$$

Step 5: Equilibrium temperature depression. At equilibrium, $P_{\text{cool}} = 0$. Linearizing for small ΔT:

$$0 = P_{\text{cool}}(300) - (4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T^3 + h_{\text{conv}})\,\Delta T \tag{09.17}$$

The effective total conductance per unit area is:

$$g_{\text{total}} = 4 \times 0.95 \times 5.67 \times 10^{-8} \times 300^3 + 5 = 5.53 + 5 = 10.53 \text{ W m}^{-2}\text{K}^{-1} \tag{09.18}$$

$$\boxed{\Delta T = \frac{P_{\text{cool}}(300)}{g_{\text{total}}} = \frac{65.0}{10.53} = 6.2\ ^\circ\text{C}} \tag{09.19}$$

Step 6: Convection hard gate check (DR-09.7).

$$h_{\text{crit}} = \frac{65.0}{6.2} - 5.53 = 10.53 - 5.53 = 5.0 \text{ W m}^{-2}\text{K}^{-1} \tag{09.20}$$

Since $h_{\text{conv}} = 5.0 \text{ W m}^{-2}\text{ K}^{-1}$ equals h_{crit} exactly, the system is operating at the convection gate boundary. Any increase in wind speed will prevent sub-ambient cooling of 6.2 °C.

Step 7: Mode-Path Matrix check. $\Gamma = G_{\text{rad}}/G_{\text{conv}} = 5.53/5 = 1.1$ (mixed-mode, confirming this is a Chapter 9 problem).

Design implication: The convective parasitic ($h_{\text{conv}} = 5 \text{ W m}^{-2}\text{ K}^{-1}$) limits ΔT to 6.2 °C. Reducing h_{conv} to 2 $\text{W m}^{-2}\text{ K}^{-1}$ (e.g., polyethylene wind shield) increases ΔT to $65/(5.53+2) = 8.6$ °C. Convection suppression is as important as spectral optimization.

Trilogy Connection: Thin-Film Spectral Optimization

The spectral emissivity profile of the As_2S_3 metasurface is a thin-film design problem. Book 1 (*The Eikonal Bridge*, Ch. 6) covers Rugate/multilayer optimization methods; Book 2 (*Quantum Field Imaging*, Ch. 8) provides the noise budget framework that analogizes to the parasitic loss budget here. The connection is that maximizing signal-to-noise in an imaging system is formally equivalent to maximizing $P_{\text{cool}}/P_{\text{sol}}$ in a radiative cooler: both require spectral shaping to suppress unwanted contributions.

09.2 Material Systems for Radiative Cooling

Three classes of material systems have demonstrated daytime sub-ambient radiative cooling. They differ in spectral selectivity, manufacturing scalability, and cost. Table 09.4 provides an overview; Table 09.5 gives the quantitative figure-of-merit comparison.

09.2.1 Polymer Films

The simplest radiative coolers use polymer films (PDMS, PMMA, or polymer–nanoparticle composites) on a metallic reflector (typically Ag or Al). The polymer provides broadband emission in the thermal IR (due to molecular vibrational modes), while the metal substrate reflects solar radiation.

Performance. PDMS/Ag films achieve $\varepsilon_{\text{eff}} \approx 0.85$ averaged over the 8–13 μm window and $\alpha_{\text{sol}} \approx 0.08$. The randomized glass-polymer hybrid metamaterial of Zhai *et al.* [2] achieves $\varepsilon_{\text{eff}} \approx 0.93$ with $\alpha_{\text{sol}} = 0.04$, demonstrating that nanostructured polymers can approach photonic-crystal performance.

Advantages. Low cost (\$2–5/m^2), roll-to-roll compatible, mechanically flexible.

Limitations. Broad emission profile extends outside the atmospheric window, increasing P_{atm}. UV degradation limits outdoor lifetime to 2–5 years without encapsulation.

09.2.2 Photonic Multilayer Stacks

Photonic multilayer stacks (e.g., SiO_2/Si_3N_4/Ag) achieve sharper spectral selectivity through thin-film interference. The seminal demonstration by Raman *et al.* [1] used a 7-layer HfO_2/SiO_2 stack on Ag, achieving sub-ambient cooling of $\sim 5\,^\circ$C under direct sunlight.

Performance. $\varepsilon_{\text{eff}} > 0.92$ in the 8–13 μm window with sharper band edges than polymer films. Out-of-band emission $\varepsilon < 0.05$ is achievable with optimized layer thicknesses. $\alpha_{\text{sol}} < 0.05$.

Advantages. Precise spectral control via layer-count and thickness optimization (Book 1, Ch. 6). Excellent UV/thermal stability (inorganic materials).

Limitations. Vacuum deposition required; cost \$20–50/m^2. Sensitive to layer thickness errors (± 2 nm tolerance for optimal performance).

09.2.3 Chalcogenide Metasurfaces

Chalcogenide glasses (As_2S_3, As_2Se_3) provide the ideal refractive index ($n \approx 2.4$–2.8) for resonant metasurface structures in the 8–13 μm atmospheric window. Patterned as nanodisk arrays on a metallic ground plane, they achieve near-ideal spectral selectivity.

Performance. $\varepsilon_{\text{eff}} > 0.95$ in the window with $\varepsilon < 0.02$ outside. $\alpha_{\text{sol}} < 0.04$ (sub-bandgap transparency above ~ 0.6 μm combined with metal reflector).

Advantages. Near-ideal spectral profile. Compatible with precision glass molding (PGM) for scalable replication (DR-6.3). Thermally stable to $> 200\,^\circ$C.

Limitations. Current cost \$50–100/m^2 for nanoimprint replication. Requires environmental encapsulation (moisture sensitivity of As_2S_3). Toxicity considerations for As-containing compositions (mitigated by sealed encapsulation).

DR-09.1 — Chalcogenide for LWIR Radiative Cooling

For LWIR radiative cooling applications requiring $\varepsilon_{\text{eff}} > 0.95$ in the 8–13 μm atmospheric window, chalcogenide glasses (As_2S_3, As_2Se_3) provide the ideal refractive index $n \approx 2.4$–2.8 for metasurface resonators. Combined with an Al or Ag ground plane, they achieve near-rectangular spectral emissivity and $\alpha_{\text{sol}} < 0.04$. For cost-sensitive applications ($<$ \$10/m^2), use the glass-polymer hybrid of Table 09.5 as the baseline.

09.2.4 Spectral Emissivity Comparison

Figure 09.4 compares the spectral emissivity profiles of the three material systems.

Figure 09.4: Spectral Emissivity of Radiative Cooling Materials

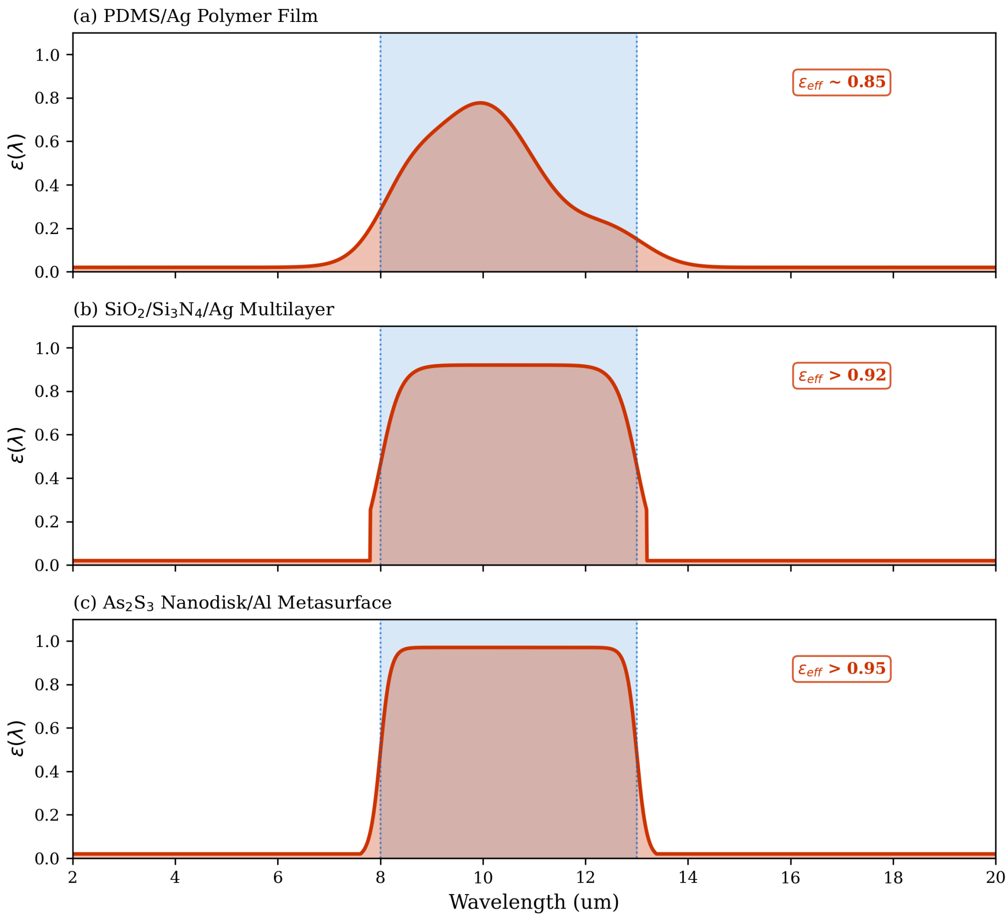


Figure 09.4: Spectral emissivity comparison of radiative cooling material systems. **(a)** PDMS/Ag polymer film: broad but incomplete coverage of the atmospheric window. **(b)** SiO_2/Si_3N_4/Ag multilayer: improved selectivity with sharper band edges. **(c)** As_2S_3 nanodisk/Al metasurface: near-ideal spectral profile with $\varepsilon > 0.95$ in the 8–13 μm window and $\varepsilon < 0.05$ outside. Blue shading marks the atmospheric transparency window.

09.2.5 Quantitative Figure-of-Merit Comparison

Table 09.5 provides the quantitative comparison that enables engineering material selection. The figure of merit $\mathrm{FOM_{cool}}$ balances cooling performance against cost and durability:

$$\mathrm{FOM_{cool}} = \frac{P_{\mathrm{cool}} \times \tau_{\mathrm{life}}}{C_{\mathrm{mfg}}} \quad [\mathrm{W\,yr\,m^{-2}\,\$^{-1}}] \tag{09.21}$$

where P_{cool} is the net cooling power at $T_{\mathrm{surf}} = T_{\mathrm{atm}}$ ($\mathrm{W\,m^{-2}}$), τ_{life} is the estimated outdoor lifetime (years), and C_{mfg} is the manufacturing cost ($\$/\mathrm{m}^2$). This FOM captures the engineering trade-off: a slightly lower-performing material with 10× lower cost and 5× longer lifetime may be preferable for large-area deployments.

Table 09.4: Material systems for radiative cooling: performance summary.

Material System	$\varepsilon_{\mathrm{eff}}$	$\varepsilon_{\mathrm{out}}$	α_{sol}	Ref.
PDMS/Ag polymer film	0.85	0.15	0.08	[11,13]
Glass-polymer hybrid	0.93	0.05	0.04	[2]
SiO_2/Si_3N_4/Ag stack	0.92	0.05	0.05	[1,12]
HfO_2/SiO_2/Ag stack	0.94	0.03	0.03	[1]
As_2S_3 nanodisk/Al	0.95	0.02	0.04	[10]

Table 09.5: Quantitative figure of merit for radiative cooling material systems. P_{cool} computed at $T_{\mathrm{surf}} = T_{\mathrm{atm}} = 300$ K, clear sky, $I_{\mathrm{sol}} = 1000\ \mathrm{W\,m^{-2}}$, no convection. $\mathrm{FOM_{cool}} = P_{\mathrm{cool}} \times \tau_{\mathrm{life}}/C_{\mathrm{mfg}}$. Higher FOM is better.

Material	P_{cool} (W/m^2)	τ_{life} (yr)	C_{mfg} (\$/m^2)	FOM (W·yr/\$)	Best Use
PDMS/Ag	40	3	3	40	Low-cost, large area
Glass-polymer hybrid	58	10	5	116	Scalable high-perf.
SiO_2/Si_3N_4/Ag	55	15	30	28	Durable, moderate area
HfO_2/SiO_2/Ag	62	15	50	19	Highest P_{cool}
As_2S_3/Al metasurf.	65	10	75	8.7	Lab/space, max ΔT

Key observations from Table 09.5. The glass-polymer hybrid dominates on FOM despite being neither the cheapest nor the highest-performing material, because it optimizes the *product* of performance, durability, and cost. The chalcogenide metasurface achieves the highest P_{cool} but its high cost makes it most suitable for applications where cooling power per unit area (not cost) is the critical metric—for example, spacecraft thermal management or high-value semiconductor equipment. For large-area terrestrial deployments (building roofs, solar panel cooling), the glass-polymer hybrid or even the simple PDMS/Ag film may be the engineering-optimal choice.

Multiphysics Integration

Material selection feeds the MPM. The material choice determines $\varepsilon_{\mathrm{eff}}$ and α_{sol}, which set G_{rad} for the surface-to-sky path in the Mode-Path Matrix. A change from PDMS/Ag ($\varepsilon_{\mathrm{eff}} = 0.85$) to As_2S_3/Al ($\varepsilon_{\mathrm{eff}} = 0.95$) increases G_{rad} by $0.95/0.85 = 1.12\times$, shifting Γ from $0.85 \times 5.53/0.95 = 4.95$ to $5.53/5 = 1.1$ (the exact value depends on h_{conv}). Both remain in the mixed-mode band, but the higher $\varepsilon_{\mathrm{eff}}$ pushes the system toward radiation dominance, making the convective design (§09.1.4, DR-09.7) even more critical.

09.3 Radiation Shielding

Radiation shielding is the complementary problem to radiative cooling: instead of maximizing radiative exchange with a cold sink, the goal is to *minimize* radiative heat transfer between two objects. In the Mode-Path Matrix, a radiation shield modifies G_{rad} by reducing the effective emissivity or inserting additional thermal resistances in series.

Recipe 09.R1 — Radiative Cooler → MPM Edge

Inputs: cooler ε_{eff} (8–13 μm), α_{sol}, area A, ambient T_{atm}, h_{conv}.

Procedure:

1. Compute radiation conductance (surface → sky): $G_{\text{rad}} = 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T_{\text{atm}}^3 A$.
2. Compute convection conductance (surface → air): $G_{\text{conv}} = h_{\text{conv}}A$.
3. Solar loading enters as a fixed source: $Q_{\text{sol}} = \alpha_{\text{sol}} I_{\text{sol}} A$.
4. Insert into MPM as two parallel edges (rad and conv) from the surface node to the environment node, plus a source term at the surface node.
5. Compute $\Gamma = G_{\text{rad}}/G_{\text{conv}}$. If $0.1 < \Gamma < 10$: mixed-mode → use Algorithm 09.A (§09.5.3).

Output: MPM edges with G_{rad}, G_{conv}, Q_{sol}, and Γ classification.

09.3.1 Single-Shield Analysis

A single radiation shield (a low-emissivity surface) placed between two parallel plates at temperatures T_1 and T_2 reduces the net radiative exchange. For gray, diffuse surfaces with emissivities ε_1, ε_2, and shield emissivity ε_s on both sides:

$$q_{\text{shielded}} = \frac{\sigma_{\text{SB}}(T_1^4 - T_2^4)}{\left(\dfrac{1}{\varepsilon_1} + \dfrac{1}{\varepsilon_2} - 1\right) + \left(\dfrac{2}{\varepsilon_s} - 1\right)} \tag{09.22}$$

The attenuation factor relative to the unshielded case is:

$$\mathcal{A}_1 = \frac{q_{\text{shielded}}}{q_{\text{unshielded}}} = \frac{(1/\varepsilon_1 + 1/\varepsilon_2 - 1)}{(1/\varepsilon_1 + 1/\varepsilon_2 - 1) + (2/\varepsilon_s - 1)} \tag{09.23}$$

For a polished aluminum shield ($\varepsilon_s = 0.05$) between two surfaces with $\varepsilon_1 = \varepsilon_2 = 0.9$:

$$\mathcal{A}_1 = \frac{(1/0.9 + 1/0.9 - 1)}{(1/0.9 + 1/0.9 - 1) + (2/0.05 - 1)} = \frac{1.22}{1.22 + 39} = 0.030 \tag{09.24}$$

A single low-ε shield reduces radiative transfer by a factor of 33.

09.3.2 Multi-Shield Networks

For N identical shields, each with emissivity ε_s:

$$\mathcal{A}_N = \frac{1}{1 + N(2/\varepsilon_s - 1)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)} \tag{09.25}$$

Each additional shield adds the same thermal resistance in series. In the Mode-Path Matrix, N shields multiply the radiation path resistance by approximately $(N + 1)$:

$$G_{\text{rad}}^{\text{shielded}} \approx \frac{G_{\text{rad}}}{N + 1} \qquad (\text{for } \varepsilon_s \ll 1) \tag{09.26}$$

DR-09.2 — Radiation Shield Count

For a target radiation attenuation factor $\mathcal{A}$, the minimum number of shields is:

$$N \geq \frac{(1-\mathcal{A})}{\mathcal{A}} \cdot \frac{(1/\varepsilon_1 + 1/\varepsilon_2 - 1)}{(2/\varepsilon_s - 1)} \tag{DR-09.2}$$

For $\varepsilon_s = 0.05$ and equal surface emissivities $\varepsilon = 0.9$, achieving $\mathcal{A} < 0.01$ requires $N \geq 3$ shields. Each shield must be mechanically supported without creating conduction short-circuits (DR-09.4).

In the MPM, each shield replaces a single $G_{\rm rad}$ element with a cascade of $(N+1)$ resistances in series plus N intermediate temperature nodes. The shield temperatures $T_{s,1}, \ldots, T_{s,N}$ become unknowns in the network and must be solved self-consistently.

Recipe 09.R2 — Radiation Shield → MPM Edge

Inputs: unshielded $G_{\rm rad}$, shield count N, shield emissivity ε_s, surface emissivities ε_1, ε_2.

Procedure:

1. Compute attenuation factor $\mathcal{A}_N$ via Eq. (09.25).
2. Compute shielded conductance: $G_{\rm rad}^{\rm shielded} = \mathcal{A}_N \times G_{\rm rad,unshielded}$.
3. In the MPM, *replace* the original $G_{\rm rad}$ edge with:
   - $(N+1)$ series radiation resistances ($1/G_{{\rm rad},k}$ each), or
   - A single equivalent edge with $G = G_{\rm rad}^{\rm shielded}$ (lumped model; valid when shield temperatures are not needed).
4. Add N intermediate temperature nodes if shield temperatures are required for thermal stress analysis.
5. Re-compute Γ for the shielded path. Shielding reduces $G_{\rm rad}$, which *decreases* Γ, potentially shifting the path from mixed-mode into the conduction-dominated regime.

Output: Updated MPM with shielded $G_{\rm rad}$ edge(s) and revised Γ.

Radiation Mode (Thermal Photonics)

Radiation shielding in cryogenic systems. Dilution refrigerators (Chapter 14) use 4–6 radiation shields between the 4 K stage and the millikelvin mixing chamber. At cryogenic temperatures, $G_{\rm rad} \propto T^3$ becomes extremely small, but so does the cooling power (20 μW at 20 mK). Even sub-nanowatt radiative loads through imperfect shields are significant. The low-emissivity coatings and IR-blocking filters used in cryostats are, in effect, the cold mirrors and spectral elements of Chapter 8 optimized for far-infrared rejection.

09.3.3 Spectral Shielding

A spectrally selective shield can provide different attenuation in different wavelength bands. This is valuable when optical access is needed (e.g., pyrometry, alignment) but thermal radiation must be blocked:

$$\varepsilon_{\text{eff}}^{\text{shield}} = \frac{\int_{\text{blocked}} \varepsilon_s(\lambda)\, B(\lambda, T)\, \mathrm{d}\lambda}{\int_0^\infty B(\lambda, T)\, \mathrm{d}\lambda} \tag{09.27}$$

The cold mirrors of Chapter 8 (§08.3) serve precisely this function: they are simultaneously radiation shields (low ε in the thermal IR) and optical windows (high transmittance at diagnostic wavelengths).

DR-09.3 — Spectral Shield Specification

When both thermal shielding and optical access are required, specify the shield as a cold mirror (Chapter 8, DR-10.3) with: (a) $\varepsilon < 0.1$ across the thermal band ($\lambda > 5$ μm), (b) $\tau > 0.9$ at the diagnostic wavelength, and (c) thermal cycling stability per DR-10.7. Compute the effective G_{rad} using ε_{eff} from Eq. (09.27) and update the MPM via Recipe 09.R2.

09.4 Multiphysics Cloaking

Thermal cloaking (Chapter 5) makes an object invisible to thermal observation by restoring the background temperature gradient. But if an observer uses an IR camera, they detect *radiation*, not conduction. A conduction-only cloak (Part II) succeeds in making the temperature field uniform outside the cloak, but the cloaked object may still emit detectable radiation if its emissivity differs from the background. True thermal invisibility requires *multiphysics cloaking*: simultaneous control of both conduction and radiation pathways.

09.4.1 The Dual-Mode Challenge

Consider a hot electronic component embedded in a substrate that must appear thermally invisible to both a contact temperature sensor and an IR camera. The Mode-Path Matrix for this system has (at minimum) two parallel paths from the component to the observer:

1. **Conduction path:** Heat conducts through the substrate to the surface, where the contact sensor measures temperature. Cloaking requires the Part II (diffusionics) approach: shape $\boldsymbol{\kappa}$ to restore the background gradient.
2. **Radiation path:** The component and surrounding surfaces emit thermal radiation toward the IR camera. Cloaking requires the Part III (spectral engineering) approach: match the surface emissivity of the cloak to the background.

The multiphysics cloaking efficiency combines both modes:

$$R_{\text{multi}} = R_{\text{cloak,cond}} \times R_{\text{cloak,rad}} \tag{09.28}$$

where $R_{\text{cloak,cond}}$ is the conduction cloaking efficiency (Chapter 5, Eq. (05.1)) and $R_{\text{cloak,rad}}$ measures the radiation-mode invisibility, defined rigorously in Metric 09.M below.

09.4.2 Instrument-Consistent Radiation Cloaking Metric

A key weakness identified in the v1 manuscript was that $R_{\text{cloak,rad}}$ was defined via an unweighted L2 norm of the emissivity difference, leaving the spectral band, weighting function, and angular treatment unspecified. This made the metric mathematically clean but experimentally ambiguous. The following definition resolves all three ambiguities.

Metric 09.M — Instrument-Consistent Radiation Cloaking Efficiency

Definition. The radiation cloaking efficiency $R_{\text{cloak,rad}}$ quantifies how closely the cloaked surface matches the background in the spectral band and angular range relevant to the detection instrument:

$$\boxed{R_{\text{cloak,rad}} = 1 - \frac{\displaystyle\int_{\lambda_1}^{\lambda_2} \left|\varepsilon_{\text{cloak}}(\lambda) - \varepsilon_{\text{bg}}(\lambda)\right| W(\lambda, T_{\text{surf}})\,\mathrm{d}\lambda}{\displaystyle\int_{\lambda_1}^{\lambda_2} \varepsilon_{\text{bg}}(\lambda)\,W(\lambda, T_{\text{surf}})\,\mathrm{d}\lambda}} \tag{09.29}$$

Spectral band. $[\lambda_1, \lambda_2]$ is the detection band:

- LWIR thermal camera: $\lambda_1 = 8\ \mu$m, $\lambda_2 = 14\ \mu$m (default).
- MWIR camera: $\lambda_1 = 3\ \mu$m, $\lambda_2 = 5\ \mu$m.
- Custom instrument: use the instrument's spectral response range.

Weighting function. $W(\lambda, T)$ specifies how spectral differences are weighted:

- *Planck-weighted* (default for thermal analysis): $W(\lambda, T) = B(\lambda, T)$, the Planck spectral radiance at the surface temperature. This weights emissivity differences near the emission peak most heavily.
- *Camera-weighted* (for signature assessment): $W(\lambda, T) = D(\lambda)\,B(\lambda, T)$, where $D(\lambda)$ is the detector spectral responsivity.
- *Flat-weighted* (for material characterization): $W(\lambda, T) = 1$.

Angular treatment. The default assumes Lambertian (diffuse) surfaces. For non-Lambertian surfaces, replace $\varepsilon(\lambda)$ with the hemispherically averaged directional emissivity:

$$\bar{\varepsilon}(\lambda) = 2\int_0^{\pi/2} \varepsilon(\lambda, \theta)\,\cos\theta\,\sin\theta\,\mathrm{d}\theta \tag{09.30}$$

For specular surfaces or oblique observation angles ($\theta > 60°$), the directional emissivity $\varepsilon(\lambda, \theta_{\text{obs}})$ should be used directly, and this must be stated in the analysis.

Interpretation. $R_{\text{cloak,rad}} = 1$ means perfect radiation invisibility in the specified band. $R_{\text{cloak,rad}} = 0$ means the cloak is fully detectable. For engineering purposes, $R_{\text{cloak,rad}} > 0.90$ is typically required for effective camouflage (Chapter 12).

Multiphysics Integration

Why separate solutions fail. A Part II conduction cloak uses layered composites (Ge/As_2Se_3, Chapter 5) to achieve $R_{\text{cloak,cond}} > 0.9$. But these layers have different emissivities from the background substrate. The Ge layer ($\varepsilon \approx 0.2$ at LWIR) is much less emissive than Si ($\varepsilon \approx 0.65$) or SiO_2 ($\varepsilon \approx 0.9$), creating a radiation-mode signature. Similarly, a Part III radiation shield (low-ε coating) blocks IR detection but disrupts the temperature field. *Both problems must be solved simultaneously,* and the solutions must be compatible.

Figure 09.5 illustrates the dual-mode cloaking concept.

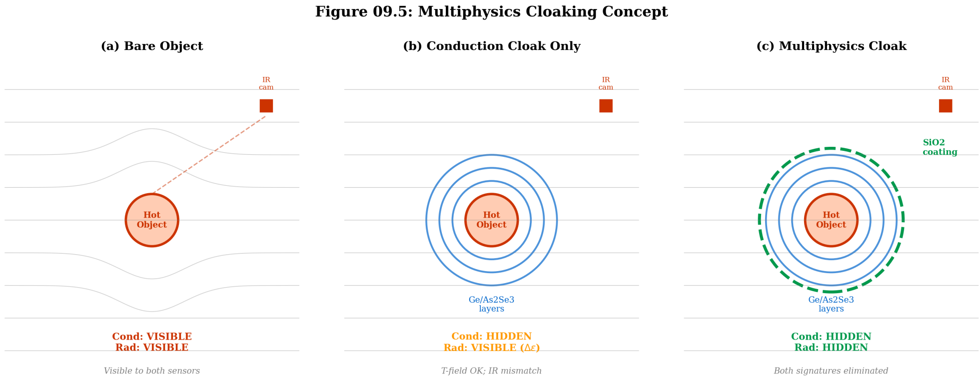


Figure 09.5: Multiphysics cloaking schematic. **(a)** Bare object: visible to both contact temperature measurement (conduction signature) and IR camera (radiation signature). **(b)** Conduction-only cloak (Part II): temperature field restored, but emissivity mismatch creates IR signature ($R_{\text{cloak,rad}} < 1$). **(c)** Multiphysics cloak: conduction cloak layers with emissivity-matched coating eliminate both signatures simultaneously ($R_{\text{multi}} \to 1$).

09.4.3 Emissivity Matching Strategies

Two approaches achieve emissivity matching:

Strategy 1: Emissivity coating on cloak layers. Apply a thin ($< 1\ \mu$m) surface coating with $\varepsilon_{\text{coat}} = \varepsilon_{\text{bg}}$. The coating must be:

- Thin enough to not perturb the conduction cloak performance (Chapter 5, sensitivity analysis: $\delta R_{\text{cloak,cond}} < 0.01$ for $t_{\text{coat}} < 2\ \mu$m).
- Thermally stable at operating temperatures.
- Fabricable using deposition methods compatible with the cloak layers (PVD, ALD, or sputtering).

SiO_2 thin films (ALD or PECVD, $\varepsilon \approx 0.9$ at LWIR) on Ge/As_2Se_3 cloak layers provide emissivity matching to a SiO_2-coated Si background. The thin SiO_2 layer adds negligible thermal resistance ($R = t/\kappa = 10^{-6}/1.4 \approx 10^{-6}\ \ \mathrm{K\,m^2/W}$).

Strategy 2: Intrinsic emissivity design. Select cloak layer materials whose intrinsic emissivity already matches the background. This is material-pair dependent and constrains the anisotropy achievable (Chapter 5, Table 05.4). For example, SiO_2/Si_3N_4 ($\varepsilon \approx 0.85$–0.90) naturally matches many oxidized silicon surfaces, but the achievable anisotropy ratio ($\kappa_\parallel/\kappa_\perp < 5$) is lower than for Ge/$As_2Se_3$ pairs.

09.4.4 Cloaking Sensitivity: Why κ-Only Optimization Fails

To quantify the coupling penalty, consider two cloak designs for a hot object on a Si substrate ($\varepsilon_{\text{bg}} = 0.65$ at LWIR):

Design A: Conduction-only (κ-optimized, no emissivity coating).

- Ge/As_2Se_3 binary cloak (Chapter 5): $R_{\text{cloak,cond}} = 0.95$.
- Cloak surface emissivity: $\varepsilon_{\text{cloak}} \approx 0.25$ (Ge-dominated surface at LWIR).
- Emissivity mismatch: $\Delta\varepsilon = |0.25 - 0.65| = 0.40$.

- Radiation cloaking (Planck-weighted, 8–14 μm): $R_{\text{cloak,rad}} \approx 1 - 0.40/0.65 = 0.38$.
- **Combined:** $R_{\text{multi}} = 0.95 \times 0.38 = 0.36$.

Design B: Multiphysics (emissivity-matched).

- Same Ge/As_2Se_3 binary cloak with 500 nm SiO_2 emissivity coating: $R_{\text{cloak,cond}} = 0.92$ (slight reduction due to coating thermal conductance).
- Cloak surface emissivity: $\varepsilon_{\text{cloak}} \approx 0.88$ (SiO_2 coating dominates at LWIR).
- Emissivity mismatch: $\Delta\varepsilon = |0.88 - 0.65| = 0.23$.
- Radiation cloaking: $R_{\text{cloak,rad}} \approx 1 - 0.23/0.65 = 0.65$.
- **Combined:** $R_{\text{multi}} = 0.92 \times 0.65 = 0.60$.

Design C: Multiphysics with optimized emissivity matching.

- SiO_2/Si_3N_4 binary cloak (intrinsic emissivity match): $R_{\text{cloak,cond}} = 0.85$ (lower anisotropy available).
- Cloak surface emissivity: $\varepsilon_{\text{cloak}} \approx 0.62$.
- Emissivity mismatch: $\Delta\varepsilon = |0.62 - 0.65| = 0.03$.
- Radiation cloaking: $R_{\text{cloak,rad}} \approx 0.95$.
- **Combined:** $R_{\text{multi}} = 0.85 \times 0.95 = 0.81$.

Table 09.6: Cloaking sensitivity comparison: three design strategies.

Design	$R_{\text{cloak,cond}}$	$\Delta\varepsilon$	$R_{\text{cloak,rad}}$	R_{multi}	Verdict
A: κ-only	0.95	0.40	0.38	0.36	Poor
B: κ + coating	0.92	0.23	0.65	0.60	Moderate
C: Intrinsic match	0.85	0.03	0.95	0.81	Best R_{multi}

Figure 09.6 plots R_{multi} contours in the ($R_{\text{cloak,cond}}, R_{\text{cloak,rad}}$) plane, showing that optimizing $R_{\text{cloak,cond}}$ alone (moving right in the plot) without addressing $R_{\text{cloak,rad}}$ (moving up) yields diminishing returns. The optimal design trajectory is diagonal, trading modest conduction performance for significant radiation improvement.

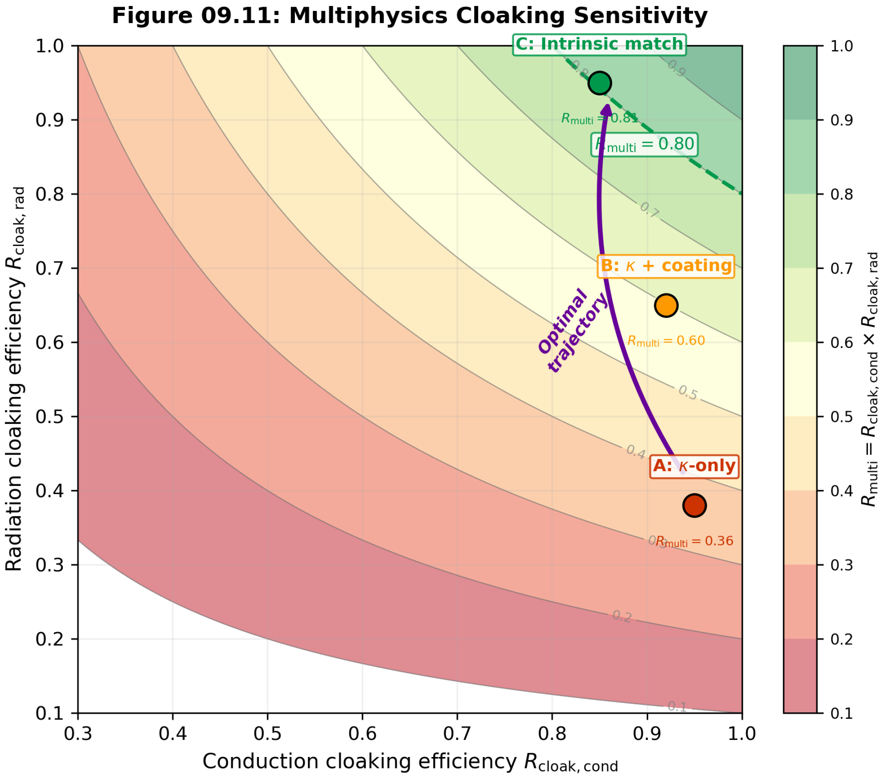


Figure 09.6: Multiphysics cloaking sensitivity. Contours show $R_{\text{multi}} = R_{\text{cloak,cond}} \times R_{\text{cloak,rad}}$ in the $(R_{\text{cloak,cond}}, R_{\text{cloak,rad}})$ plane. Design A (κ-only) lies in the lower-right corner; Design C (intrinsic match) approaches the upper diagonal. The dashed line marks $R_{\text{multi}} = 0.80$, a typical engineering threshold for effective camouflage. The arrow indicates the optimal design trajectory: trading modest $R_{\text{cloak,cond}}$ reduction for large $R_{\text{cloak,rad}}$ gain.

09.4.5 Multiphysics Compatibility Requirements

DR-09.4 — Multiphysics Cloaking Compatibility

When designing a multiphysics cloak, ensure:

(a) Conduction cloak layers (Part II) do not create radiation-mode signatures: $\Delta\varepsilon < 0.05$ at LWIR (Planck-weighted per Metric 09.M).

(b) Emissivity coatings do not short-circuit the conduction design: $t_{\text{coat}} < 2\ \mu$m, $\kappa_{\text{coat}} \cdot t_{\text{coat}} < 10^{-3}$ W/K per unit area.

(c) Shield supports do not create conduction bridges between hot and cold surfaces (use low-κ standoffs: PEEK, aerogel, glass fiber).

(d) The combined efficiency $R_{\text{multi}} = R_{\text{cloak,cond}} \times R_{\text{cloak,rad}} > 0.80$ for effective camouflage applications (Chapter 12).

09.4.6 Worked Example: WE-09.2—Multiphysics Cloak Assessment

Problem: A Ge/As_2Se_3 binary cloak (Chapter 5) is used on a Si substrate to hide a heat source. Assess whether an emissivity coating is needed and compute R_{multi} using Metric 09.M.

Given:

- Conduction cloak: $R_{\text{cloak,cond}} = 0.92$ (Chapter 5, Table 05.5).

- Cloak surface: Ge layer ($\varepsilon_{\text{Ge}} \approx 0.20$ at 10 μm LWIR).
- Background: bare Si ($\varepsilon_{\text{bg}} = 0.65$ at LWIR).
- Surface temperature: $T_{\text{surf}} = 350$ K.
- Detection: LWIR thermal camera ($\lambda = 8$–14 μm).

Step 1: Compute $R_{\text{cloak,rad}}$ without coating.
Using Metric 09.M with Planck weighting at 350 K over 8–14 μm (gray-surface approximation: emissivity independent of λ within the band):

$$R_{\text{cloak,rad}} = 1 - \frac{|\varepsilon_{\text{Ge}} - \varepsilon_{\text{bg}}| \int_8^{14} B(\lambda, 350)\, \mathrm{d}\lambda}{\varepsilon_{\text{bg}} \int_8^{14} B(\lambda, 350)\, \mathrm{d}\lambda} = 1 - \frac{|0.20 - 0.65|}{0.65} = 1 - 0.692 = 0.31 \quad (09.31)$$

Step 2: Combined efficiency without coating.

$$R_{\text{multi}} = 0.92 \times 0.31 = 0.28 \quad (09.32)$$

This is far below the $R_{\text{multi}} > 0.80$ threshold (DR-09.4d). *The conduction-only cloak is radiatively conspicuous.*

Step 3: Apply SiO_2 emissivity coating (500 nm ALD).

- New surface emissivity: $\varepsilon_{\text{coat}} = 0.88$ (SiO_2 at LWIR).
- Conduction penalty: $\delta R_{\text{cloak,cond}} = -0.01$ (from Chapter 5 sensitivity analysis for 500 nm SiO_2 on Ge).
- Updated: $R_{\text{cloak,cond}} = 0.92 - 0.01 = 0.91$.

$$R_{\text{cloak,rad}} = 1 - \frac{|0.88 - 0.65|}{0.65} = 1 - 0.354 = 0.65 \quad (09.33)$$

$$R_{\text{multi}} = 0.91 \times 0.65 = 0.59 \quad (09.34)$$

Still below 0.80. The $\Delta\varepsilon = 0.23$ is too large.

Step 4: Use oxidized Si background ($\varepsilon_{\text{bg}} = 0.90$) and SiO_2 coating.
Many semiconductor surfaces are thermally oxidized, giving $\varepsilon_{\text{bg}} \approx 0.90$. With SiO_2 coating ($\varepsilon_{\text{coat}} = 0.88$):

$$R_{\text{cloak,rad}} = 1 - \frac{|0.88 - 0.90|}{0.90} = 1 - 0.022 = 0.98 \quad (09.35)$$

$$\boxed{R_{\text{multi}} = 0.91 \times 0.98 = 0.89} \quad (09.36)$$

This exceeds the $R_{\text{multi}} > 0.80$ threshold.

Design lesson: The effectiveness of emissivity matching depends critically on the background surface condition. On oxidized Si ($\varepsilon = 0.90$), a simple SiO_2 ALD coating achieves excellent radiation cloaking. On bare Si ($\varepsilon = 0.65$), no simple oxide coating can match, and intrinsic material selection (Design C in Table 09.6) becomes necessary.

09.4.7 MPM Integration for Multiphysics Cloaking

Recipe 09.R3 — Multiphysics Cloak → MPM Subgraph

Inputs: cloak geometry, layer materials (κ_i, ε_i), background properties (κ_{bg}, ε_{bg}), object heat generation Q, surface temperature T_{surf}.

Procedure:

1. **Conduction subgraph:** Compute effective $\boldsymbol{\kappa}$ for the cloak shell using Chapter 5 EMT. Insert as G_{cond} edges between the object node, shell nodes, and background nodes.
2. **Radiation subgraph:** For each surface exposed to an IR observer:
   (i) Compute $\varepsilon_{\text{cloak}}(\lambda)$ (surface material or coating).
   (ii) Compute $R_{\text{cloak,rad}}$ via Metric 09.M.
   (iii) Compute $G_{\text{rad}} = 4\varepsilon_{\text{cloak}}\sigma_{\text{SB}}T_{\text{surf}}^3 A$ for each radiating surface.
3. **Coupling check:** If any cloak path has $0.1 < \Gamma < 10$, the conduction and radiation subgraphs are coupled → use Algorithm 09.A (§09.5.3).
4. **Compatibility check:** Verify DR-09.4 constraints (emissivity mismatch, coating thermal short-circuit, support conductance).
5. **Output:** R_{multi} for the system, plus the updated MPM with all conductance edges.

Output: Complete MPM subgraph for the cloaked region, with $R_{\text{cloak,cond}}$, $R_{\text{cloak,rad}}$, R_{multi}, and Γ for each path.

Trilogy Connection: Emissivity as a Design Variable

In Book 1 (*The Eikonal Bridge*), surface coatings are designed for optical performance (reflectance, transmittance). Here, the same thin-film design tools serve a thermal function: matching emissivity to a background. The design methodology is identical—layer count, material selection, thickness optimization—but the merit function changes from optical MTF to $R_{\text{cloak,rad}}$ (Metric 09.M). This is a concrete example of the trilogy's unifying principle: the same physical optics governs both imaging quality and thermal signature.

09.5 Mixed-Mode Design Methodology

This section formalizes the design methodology for the mixed-mode regime $0.1 < \Gamma < 10$, completing the tool set promised by the Mode-Path Matrix framework of Chapter 4. We begin by demonstrating *why* a self-consistent iteration is necessary (§09.5.1), then present the formal algorithm (§09.5.3), analyze its convergence (§09.5.4), and quantify the error from ignoring mode coupling (§09.5.5).

09.5.1 Why Linear Superposition Fails

The temptation in mixed-mode analysis is to simply add the conduction and radiation conductances: $G_{\text{total}} \stackrel{?}{=} G_{\text{cond}} + G_{\text{rad}}(T_0)$, where T_0 is some reference temperature. This section shows why this approach introduces systematic errors that grow with Γ.

The feedback mechanism. Consider a two-node system: node 1 (heated, temperature T_1) connected to node 2 (fixed at T_2) by parallel conduction and radiation paths. Energy balance at node 1:

$$Q = G_{\text{cond}}(T_1 - T_2) + G_{\text{rad}}(T_1)\,(T_1 - T_2) \tag{09.37}$$

where the linearized radiation conductance is:

$$G_{\text{rad}}(T_1) = 4\varepsilon F_{12}\sigma_{\text{SB}} T_m^3 A \qquad \text{with} \quad T_m = \frac{T_1 + T_2}{2} \tag{09.38}$$

In the "superposition" approach, one evaluates G_{rad} at the *initial guess* $T_1^{(0)}$ (typically the conduction-only solution) and solves:

$$T_1^{\text{super}} = T_2 + \frac{Q}{G_{\text{cond}} + G_{\text{rad}}(T_1^{(0)})} \tag{09.39}$$

But G_{rad} depends on T_1 through T_m^3. The *true* solution T_1^* satisfies:

$$Q = G_{\text{cond}}(T_1^* - T_2) + 4\varepsilon F_{12}\sigma_{\text{SB}} \left(\frac{T_1^* + T_2}{2}\right)^3 A\,(T_1^* - T_2) \tag{09.40}$$

Quantifying the superposition error. Define $\Delta T = T_1^{(0)} - T_1^*$ as the error from using the conduction-only initial guess. Since adding $G_{\text{rad}} > 0$ reduces T_1, the conduction-only solution *overestimates* T_1. To first order in $\Delta T/T$:

$$\frac{\Delta T}{T_1^* - T_2} \approx \frac{\Gamma}{1 + \Gamma} \tag{09.41}$$

where $\Gamma = G_{\text{rad}}/G_{\text{cond}}$ is evaluated at T_1^*. This is a *systematic* error: it always overestimates the temperature drop.

The deeper problem is the *feedback loop*: changing T_1 changes G_{rad} (through T_m^3), which changes T_1, which changes G_{rad} again. Define the sensitivity:

$$\frac{\partial G_{\text{rad}}}{\partial T_1} = 4\varepsilon F_{12}\sigma_{\text{SB}} A \cdot \frac{3}{2}\left(\frac{T_1 + T_2}{2}\right)^2 \cdot \frac{1}{2} = 3\varepsilon F_{12}\sigma_{\text{SB}} A\, T_m^2 \tag{09.42}$$

The feedback gain (ratio of radiation sensitivity to total conductance) is:

$$\mathcal{F} = \frac{(T_1 - T_2)}{G_{\text{cond}} + G_{\text{rad}}} \cdot \frac{\partial G_{\text{rad}}}{\partial T_1} \approx \frac{3\Gamma}{1 + \Gamma} \cdot \frac{\Delta T}{T_m} \tag{09.43}$$

When $\Gamma \sim 1$ and $\Delta T/T_m \sim 0.1$ (a 30 K drop from 300 K), $\mathcal{F} \approx 0.15$: the feedback is significant and the superposition approach fails.

Failure Mode: Superposition Error in Mixed-Mode Systems

Using $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}}(T_0)$ with a *fixed* T_0 introduces a systematic relative error $\Delta R/R = \Gamma/(1+\Gamma)$ in the thermal resistance. At $\Gamma = 1$, this is a 50% error. At $\Gamma = 5$, it is 83%. The only remedy is iterative self-consistent solution: **Algorithm 09.A**.

09.5.2 The Coupled Network

In a general thermal network with M nodes at temperatures $T_1, \ldots, T_M$, each node satisfies energy conservation:

$$\sum_{j \neq i} G_{ij}^{\text{cond}}(T_j - T_i) + \sum_{j \neq i} G_{ij}^{\text{rad}}(\mathbf{T})(T_j - T_i) + Q_i = 0 \tag{09.44}$$

where Q_i is the heat generation at node i and the radiation conductance is temperature-dependent:

$$G_{ij}^{\mathrm{rad}}(\mathbf{T}) = 4\varepsilon_{ij} F_{ij} \sigma_{\mathrm{SB}} \bar{T}_{ij}^3 A_{ij} \tag{09.45}$$

with $\bar{T}_{ij} = (T_i + T_j)/2$ being the mean temperature of the pair.

The system in matrix form is:

$$\left[\mathbf{G}^{\mathrm{cond}} + \mathbf{G}^{\mathrm{rad}}(\mathbf{T})\right] \mathbf{T} = \mathbf{Q} + \mathbf{b} \tag{09.46}$$

where $\mathbf{b}$ accounts for boundary conditions. Because $\mathbf{G}^{\mathrm{rad}}$ depends on $\mathbf{T}$, this is a *nonlinear* system that requires iterative solution.

09.5.3 Algorithm 09.A: Coupled Conduction–Radiation Solver

Algorithm 1 Algorithm 09.A — Coupled Conduction–Radiation Solver for Mixed-Mode Networks

Require: Network graph $\mathcal{G}$ with M nodes; $\mathbf{G}^{\text{cond}}$ (constant); material properties ε_{ij}, F_{ij}, A_{ij}; boundary conditions $\mathbf{b}$; heat sources $\mathbf{Q}$; tolerances ϵ_T, ϵ_E; max iterations $N_{\max}$; relaxation parameter ω

Ensure: Converged temperature vector $\mathbf{T}^*$; Γ for each path; convergence flag

S1: Initialize. Solve conduction-only: $\mathbf{G}^{\text{cond}}\,\mathbf{T}^{(0)} = \mathbf{Q} + \mathbf{b}$
Set iteration counter $k \leftarrow 0$

repeat
  $k \leftarrow k + 1$

  S2: Compute radiation conductances.
  for each edge (i, j) with radiation coupling **do**
    $\bar{T}_{ij} \leftarrow (T_i^{(k-1)} + T_j^{(k-1)})/2$
    $G_{ij}^{\text{rad}} \leftarrow 4\varepsilon_{ij} F_{ij} \sigma_{\text{SB}} \bar{T}_{ij}^3 A_{ij}$
  end for

  S3: Solve coupled system.
  Solve $[\mathbf{G}^{\text{cond}} + \mathbf{G}^{\text{rad}}(\mathbf{T}^{(k-1)})]\,\tilde{\mathbf{T}}^{(k)} = \mathbf{Q} + \mathbf{b}$

  S4: Under-relaxation.
  $\mathbf{T}^{(k)} \leftarrow \omega\,\tilde{\mathbf{T}}^{(k)} + (1-\omega)\,\mathbf{T}^{(k-1)}$

  S5: Convergence check (dual criteria).
  Temperature residual: $r_T \leftarrow \|\mathbf{T}^{(k)} - \mathbf{T}^{(k-1)}\|_\infty$
  Energy residual: $r_E \leftarrow \max_i |\sum_j (G_{ij}^{\text{cond}} + G_{ij}^{\text{rad}})(T_j^{(k)} - T_i^{(k)}) + Q_i|$

  S6: Fallback check.
  if $k > N_{\max}/2$ **and** $r_T^{(k)}/r_T^{(k-1)} > 0.9$ **then**
    Switch to Newton–Raphson with Jacobian: $J_{ij} = \partial r_i/\partial T_j$ where $\partial G_{ij}^{\text{rad}}/\partial T_j = 6\varepsilon_{ij} F_{ij} \sigma_{\text{SB}} \bar{T}_{ij}^2 A_{ij}$
    (see Note below)
  end if
until $(r_T < \epsilon_T)$ **and** $(r_E < \epsilon_E)$ **or** $k \geq N_{\max}$

S7: Post-processing.
for each edge (i, j) **do**
  $\Gamma_{ij} \leftarrow G_{ij}^{\text{rad}}/G_{ij}^{\text{cond}}$
end for

S8: Error estimation.
for each edge (i, j) **do**
  $\delta G_{ij}^{\text{rad}} \leftarrow 12\varepsilon_{ij} F_{ij} \sigma_{\text{SB}} \bar{T}_{ij}^2 A_{ij} \cdot \delta T$ ▷ From $\partial G_{\text{rad}}/\partial T$
end for

if $k \geq N_{\max}$ **then**
  return `NON-CONVERGED`, $\mathbf{T}^{(k)}$, Γ
else
  return `CONVERGED`, $\mathbf{T}^{(k)}$, Γ, $\delta\mathbf{G}^{\text{rad}}$
end if

Recommended default parameters.

- Temperature tolerance: $\epsilon_T = 0.01$ K (semiconductor specification-driven; for building-scale systems, $\epsilon_T = 0.1$ K may suffice).
- Energy residual tolerance: $\epsilon_E = 10^{-3} \times Q_{\max}$ (0.1% of the largest source term).
- Maximum iterations: $N_{\max} = 50$ (generous; convergence typically occurs in 3–15 iterations).
- Relaxation parameter: ω per Table 09.7.

Note on Newton–Raphson fallback (S6). When successive substitution converges slowly ($\rho > 0.9$), the convergence rate can be dramatically improved by switching to Newton–Raphson iteration. The Jacobian of the residual vector requires $\partial G_{ij}^{\mathrm{rad}}/\partial T_k$, which is analytically available from Eq. (09.42). For small networks ($M < 20$ nodes), direct LU factorization of the Jacobian is efficient. For larger networks, Anderson acceleration (a quasi-Newton method that does not require explicit Jacobian assembly) is recommended.

DR-09.5 — Coupled Iteration Convergence

For mixed-mode systems ($0.1 < \Gamma < 10$), use Algorithm 09.A with dual convergence criteria ($r_T < \epsilon_T$ and $r_E < \epsilon_E$). Select under-relaxation per Table 09.7: $\omega = 0.5$ for $\Gamma < 5$; $\omega = 0.3$ for $\Gamma > 5$. The algorithm converges in 3–15 iterations for well-conditioned networks. If convergence stalls after $N_{\max}/2$ iterations, switch to Newton–Raphson (S6). Flag non-convergence at $N_{\max} = 50$ and inspect the network for pathological couplings (radiation feedback loops, missing conduction paths).

Transient Regime

Transient coupled analysis. Algorithm 09.A applies to steady state. For transient problems, the coupled system becomes:

$$\mathbf{C}\,\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} + \left[\mathbf{G}^{\mathrm{cond}} + \mathbf{G}^{\mathrm{rad}}(\mathbf{T})\right]\mathbf{T} = \mathbf{Q}(t) + \mathbf{b}(t) \tag{09.47}$$

where $\mathbf{C}$ is the diagonal thermal capacitance matrix. This requires a nonlinear ODE solver (implicit schemes preferred for stiff radiation terms). Commercial tools (COMSOL, Ansys) handle this automatically; for network models, backward Euler with inner Newton–Raphson iterations at each time step is recommended. At each time step, the inner iteration is Algorithm 09.A applied to the time-discretized system.

Figure 09.7 provides a visual summary of Algorithm 09.A.

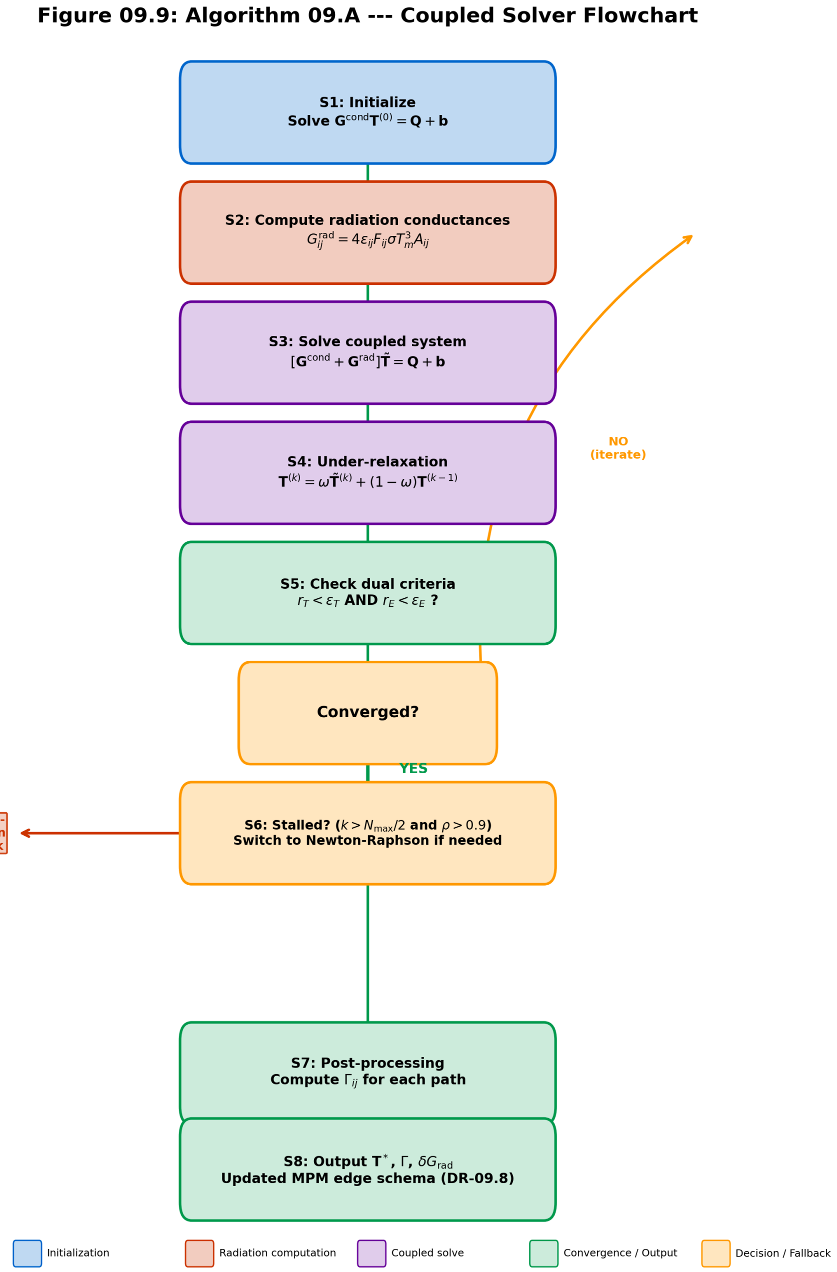


Figure 09.7: Algorithm 09.A flowchart. The coupled solver initializes with a conduction-only solution (S1), then iterates: compute $G_{\text{rad}}(T)$ (S2), solve coupled system (S3), apply under-relaxation (S4), check dual convergence criteria (S5). If convergence stalls, a Newton–Raphson fallback (S6) is triggered. Post-processing (S7) computes Γ for each path; error estimation (S8) provides δG_{rad} bounds. Green path: normal convergence. Orange path: fallback branch. Red path: non-convergence flag.

09.5.4 Convergence Analysis

The convergence rate of the successive substitution algorithm depends on Γ. Define the convergence ratio:

$$\rho = \frac{\|\mathbf{T}^{(k+1)} - \mathbf{T}^*\|}{\|\mathbf{T}^{(k)} - \mathbf{T}^*\|} \tag{09.48}$$

where $\mathbf{T}^*$ is the converged solution. For a simple two-node system:

$$\rho \approx \omega \cdot \frac{3\Gamma}{1+\Gamma} \tag{09.49}$$

Convergence requires $\rho < 1$, which gives:

$$\omega < \frac{1+\Gamma}{3\Gamma} \tag{09.50}$$

Table 09.7: Recommended under-relaxation parameter ω vs. Γ.

Γ	ω_{max}	ω_{rec}	Typical iterations	Convergence rate
0.1	3.7	0.7	3–4	Fast
0.5	1.0	0.6	4–5	Moderate
1.0	0.67	0.5	5–6	Moderate
2.0	0.50	0.4	6–8	Slow
5.0	0.40	0.3	8–12	Slow
10	0.37	0.3	10–15	Very slow

Figure 09.8 shows the convergence behavior for different Γ values.

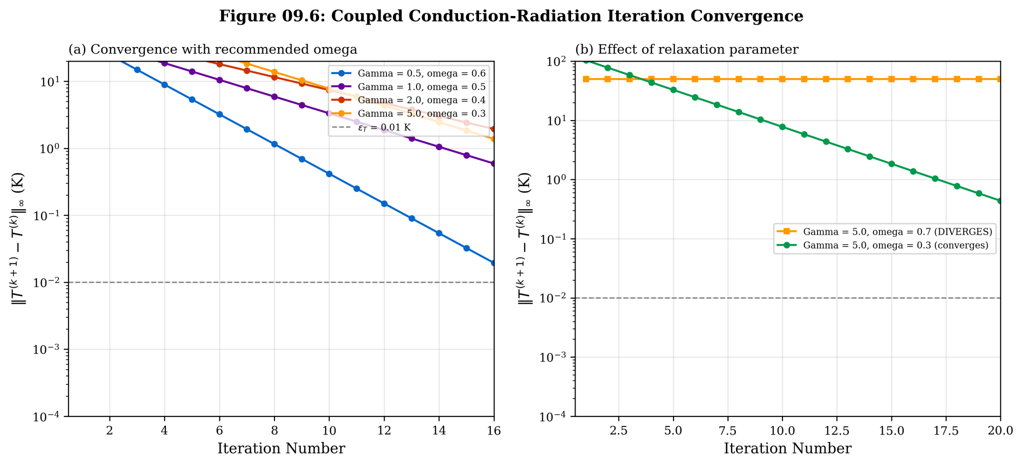


Figure 09.8: Coupled conduction–radiation iteration convergence. **(a)** Temperature residual $\|\mathbf{T}^{(k+1)} - \mathbf{T}^{(k)}\|_\infty$ vs. iteration number for $\Gamma = 0.5$, 1.0, 2.0, and 5.0 with recommended ω values. All cases converge below $\epsilon_T = 0.01$ K within 12 iterations. **(b)** Convergence failure ($\Gamma = 5.0$, $\omega = 0.7$): oscillation due to excessive relaxation parameter, demonstrating the importance of DR-09.5.

09.5.5 Error Estimation

Neglecting the radiation contribution in a mixed-mode system introduces a systematic error in temperature. For a single path with total conductance $G_{\text{total}} = G_{\text{cond}}(1+\Gamma)$, the conduction-only model predicts temperatures as if $G_{\text{total}} = G_{\text{cond}}$, introducing a relative error in thermal resistance:

$$\frac{\Delta R}{R} = \frac{\Gamma}{1+\Gamma} \tag{09.51}$$

Table 09.8: Systematic error from neglecting radiation (Γ-dependent).

Γ	$\Delta R/R$ (%)	Implication
0.01	1.0	Negligible—conduction model valid
0.1	9.1	Borderline—check specification margin
0.5	33	Significant—coupled model required
1.0	50	Large—coupled model essential
5.0	83	Dominant—radiation model with conduction correction
10	91	Near-total—radiation model valid

Figure 09.9 plots Eq. (09.51) as a continuous curve with regime boundaries, providing the visual justification for this chapter's existence.

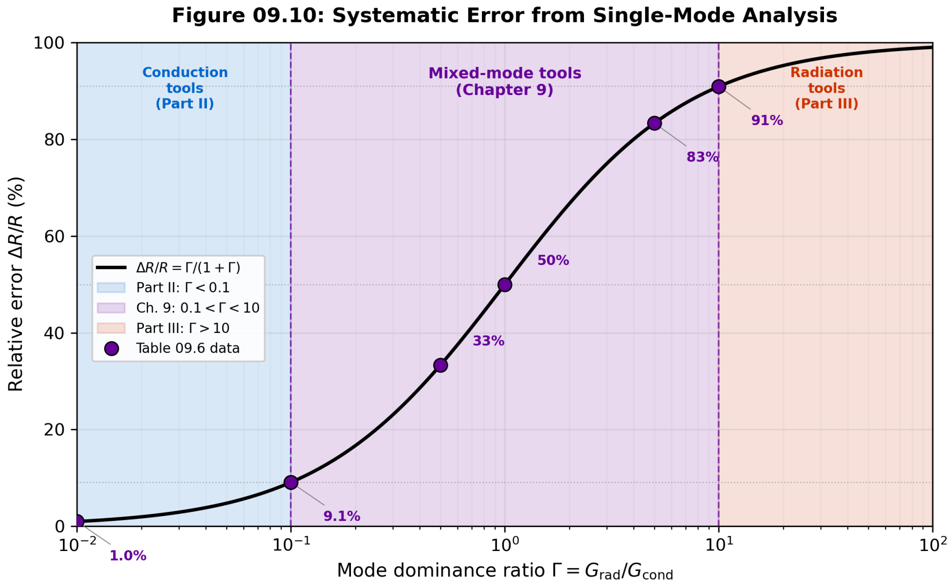


Figure 09.9: Systematic error from single-mode analysis. The curve $\Delta R/R = \Gamma/(1+\Gamma)$ shows the relative error in thermal resistance when radiation is neglected. Shaded regions indicate the three analysis regimes: blue ($\Gamma < 0.1$, Part II conduction tools), purple ($0.1 < \Gamma < 10$, this chapter's mixed-mode tools), and red ($\Gamma > 10$, Part III radiation tools). The mixed-mode band spans errors from 9% to 91%, making coupled analysis essential. Data points from Table 09.8 are overlaid.

DR-09.6 — Mixed-Mode Analysis Threshold

Use coupled conduction–radiation analysis (Algorithm 09.A) whenever any path in the Mode-Path Matrix has $0.1 < \Gamma < 10$. The systematic error from single-mode analysis exceeds 9% at $\Gamma = 0.1$ and 91% at $\Gamma = 10$. For paths near the boundaries ($\Gamma \approx 0.1$ or $\Gamma \approx 10$), the single-mode result may be used as the initial guess for Algorithm 09.A (Step S1).

Error propagation in the converged solution. After Algorithm 09.A converges, the uncertainty in G_{rad} propagates into temperature uncertainty. For a single edge with linearized radiation conductance:

$$\delta G_{\mathrm{rad}} \approx 12\varepsilon F_{12}\sigma_{\mathrm{SB}} T_m^2 A \cdot \delta T \tag{09.52}$$

where δT represents the temperature measurement or estimation uncertainty. The corresponding uncertainty in the converged node temperature is:

$$\delta T_i \approx \frac{\Gamma_i}{1+\Gamma_i} \cdot \frac{\delta G_{\text{rad}}}{G_{\text{cond}} + G_{\text{rad}}} \cdot (T_i - T_j) \tag{09.53}$$

This shows that temperature uncertainty is amplified by $\Gamma/(1+\Gamma)$—the same factor that governs the systematic error. Systems with $\Gamma \sim 1$ are most sensitive to parameter uncertainties.

Steady-State Limitation

Linearization validity. The linearized radiation conductance $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T_m^3 A$ is accurate when $|T_1 - T_2|/T_m < 0.2$, i.e., the temperature difference is less than 20% of the mean temperature. For semiconductor processes with $T_m = 500$ K, this allows ΔT up to 100 K. For larger temperature differences (e.g., furnace door to room temperature), use the full $T_1^4 - T_2^4$ formulation in Algorithm 09.A Step S2.

09.6 Semiconductor Fabrication in the Mixed-Mode Regime

Figure 09.10 maps the Γ values for common semiconductor fabrication environments, identifying those that fall in the mixed-mode band.

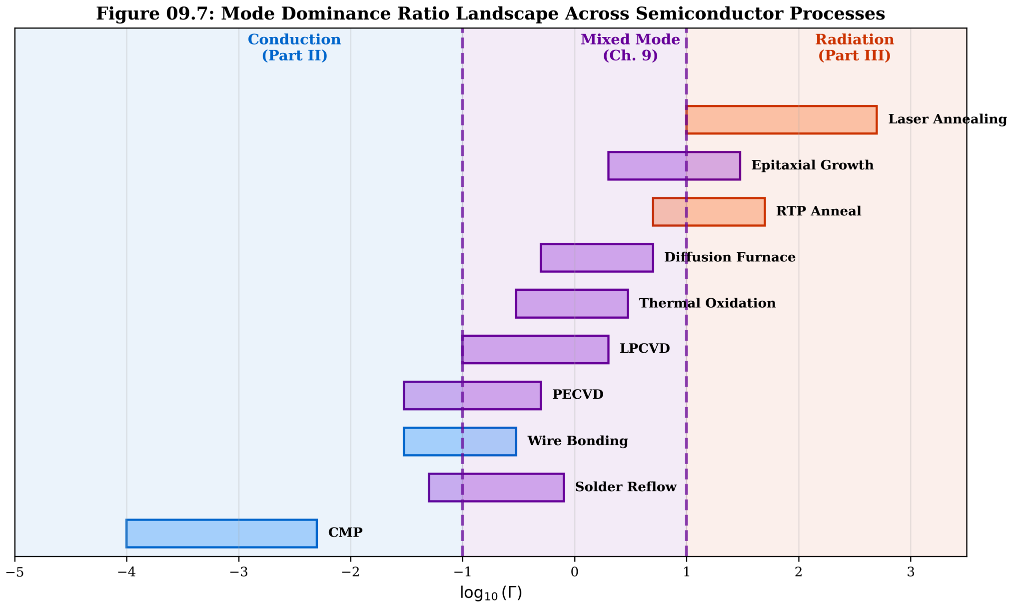


Figure 09.10: Γ landscape across semiconductor fabrication processes. Each process is mapped to its typical temperature range and dominant thermal path Γ. Processes in the purple band ($0.1 < \Gamma < 10$) require mixed-mode analysis. The PECVD and thermal oxidation regimes are highlighted as the primary targets for Algorithm 09.A.

Table 09.9: Semiconductor processes in the mixed-mode regime.

Process	**T range (°C)**	**Primary path**	**Γ**	**Regime**
PECVD (SiO_2, SiN_x)	300–400	wafer–chamber wall	0.5–3	Mixed
Thermal oxidation	800–1100	wafer–quartz tube	2–8	Mixed
RTP (spike anneal)	900–1100	wafer–lamp reflector	5–50	Mixed to rad.
LPCVD	500–700	wafer–tube wall	1–5	Mixed
Etch (plasma)	20–80	wafer–ESC	0.01–0.1	Conduction
CMP	20–40	wafer–pad	< 0.01	Conduction

The table confirms that PECVD, thermal oxidation, LPCVD, and portions of the RTP range all operate in the mixed-mode band. For these processes, conduction-only thermal models underestimate heat transfer (and thus overestimate temperature non-uniformity), while radiation-only models miss the conductive coupling through gas gaps and solid contacts.

09.6.1 Worked Example: WE-09.3—PECVD Chamber Mixed-Mode Analysis

Problem: Analyze the thermal network of a 300 mm PECVD chamber operating at 400 °C (673 K) for SiO_2 deposition. Construct the Mode-Path Matrix, identify mixed-mode paths, and solve using Algorithm 09.A.

System description (Figure 09.11):

- **Node 1: Wafer** (T_1, unknown). 300 mm diameter, Si, $\varepsilon = 0.65$. Heated by plasma ($Q_{\text{plasma}} = 800$ W total; 1130 W/m^2 over wafer area $A_w = 0.0707$ m^2).
- **Node 2: ESC (electrostatic chuck)**. Temperature-controlled at $T_2 = 673$ K. Contact conductance: $h_{\text{gap}} = 500\ \mathrm{W\,m^{-2}\,K^{-1}}$ (He backside gas).
- **Node 3: Chamber wall**. Water-cooled at $T_3 = 350$ K. $\varepsilon_{\text{wall}} = 0.5$ (anodized Al).
- **Node 4: Showerhead (gas inlet)**. Gas-heated to $T_4 = 573$ K. $\varepsilon_{\text{shower}} = 0.3$ (stainless steel). Gap to wafer: 15 mm (process gas, N_2/SiH_4 mixture, $\kappa_{\text{gas}} \approx 0.04\ \mathrm{W\,m^{-1}\,K^{-1}}$).
- **Node 5: Quartz window** (for plasma excitation). $T_5 = 473$ K (estimated). $\varepsilon_{\text{quartz}} = 0.75$. View factor to wafer: $F_{15} = 0.10$.

Step 1: Construct conduction conductances.

Path	**Mechanism**	**Expression**	**G (W/K)**
1–2	He backside gas	$h_{\text{gap}}A_w$	$500 \times 0.0707 = 35.4$
1–4	Process gas gap	$\kappa_{\text{gas}}A_w/d$	$0.04 \times 0.0707/0.015 = 0.189$

Step 2: Construct radiation conductances at $T_1^{(0)} = 673$ K (initial guess = ESC temperature).

Path	**ε_{eff}**	**F_{ij}**	**Expression**	**G_{rad} (W/K)**
1–3	0.42	0.60	$4 \times 0.42 \times 0.60 \times 5.67 \times 10^{-8} \times 512^3 \times 0.0707$	1.09
1–4	0.26	0.25	$4 \times 0.26 \times 0.25 \times 5.67 \times 10^{-8} \times 623^3 \times 0.0707$	0.32
1–5	0.54	0.10	$4 \times 0.54 \times 0.10 \times 5.67 \times 10^{-8} \times 573^3 \times 0.0707$	0.16

where the effective emissivities account for the parallel-plate configuration: $\varepsilon_{\text{eff}} = (1/\varepsilon_1 + 1/\varepsilon_j - 1)^{-1}$.

T_m values: $T_{m,13} = (673+350)/2 = 512$ K; $T_{m,14} = (673+573)/2 = 623$ K; $T_{m,15} = (673+473)/2 = 573$ K.

Step 3: Build Mode-Path Matrix.

Path	$\boldsymbol{G}_{\textbf{cond}}$ (W/K)	$\boldsymbol{G}_{\textbf{rad}}$ (W/K)	$\boldsymbol{\Gamma}$	**Mode**
1–2 (wafer–ESC)	35.4	~ 0	< 0.01	Conduction
1–3 (wafer–wall)	0 (no contact)	1.09	$\gg 10$	Radiation
1–4 (wafer–shower)	0.189	0.32	1.7	**Mixed**
1–5 (wafer–window)	0 (no contact)	0.16	$\gg 10$	Radiation

The wafer–showerhead path (1–4) is the only mixed-mode path ($\Gamma = 1.7$), requiring coupled iteration. The wafer–ESC path dominates total conductance (35.4 W/K) and is purely conductive. The wafer–wall and wafer–window paths are purely radiative.

Step 4: Apply Algorithm 09.A.

With $\omega = 0.5$ (appropriate for $\Gamma = 1.7$ per Table 09.7), $\epsilon_T = 0.01$ K, $\epsilon_E = 0.8$ W (0.1% of $Q = 800$ W):

Table 09.12: Algorithm 09.A iteration history for WE-09.3.

Iter k	$T_1^{(k)}$ (K)	$G_{\text{rad}}^{1\text{–}3}$	$G_{\text{rad}}^{1\text{–}4}$	r_T (K)	r_E (W)
0 (cond-only)	695.6	—	—	—	—
1	689.3	1.14	0.34	6.3	12.4
2	687.1	1.12	0.33	2.2	4.1
3	686.3	1.11	0.33	0.8	1.5
4	686.0	1.11	0.33	0.3	0.5
5	685.9	1.11	0.33	0.1	0.2
6	685.9	1.11	0.33	0.005	0.01

Converged result: $\boxed{T_{\text{wafer}} = 685.9\text{ K} = 412.9\,^\circ\text{C}}$ after 6 iterations.

Step 5: Error assessment.

The conduction-only prediction was $T_1 = 695.6$ K; the coupled solution is 685.9 K. The temperature *overestimate* from ignoring radiation:

$$\Delta T_{\text{error}} = 695.6 - 685.9 = 9.7\text{ K} \tag{09.54}$$

Process consequence: A 9.7 K error at 673 K is 1.4% in absolute temperature, but for a typical PECVD uniformity specification of ± 1 K, this error is *nearly 10× the specification.* The conduction-only model would predict the wafer is within spec when it is actually 10 K too hot, potentially affecting film stress, composition, and step coverage.

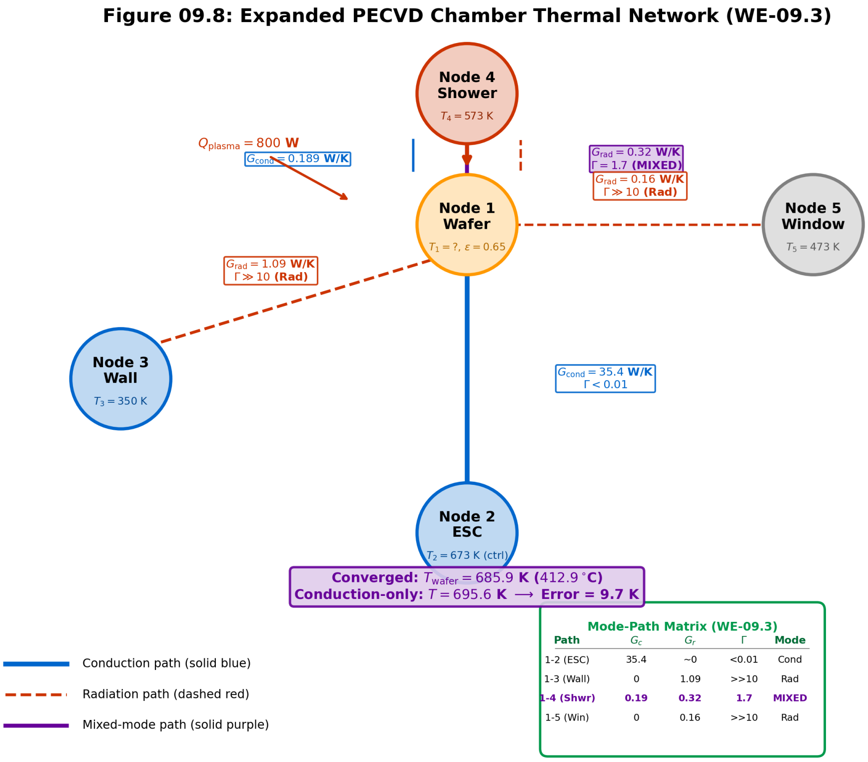
Step 6: Sensitivity analysis.

(a) **Gold-coated chamber walls** ($\varepsilon_{\text{wall}} = 0.05$): $G_{\text{rad}}^{1\text{–}3}$ drops from 1.11 to 0.11 W/K. Converged $T_1 = 694.2$ K. The system becomes nearly conduction-only ($\Gamma_{1\text{–}3} \approx 0.1$).

(b) **Enhanced He backside cooling** ($h_{\text{gap}} = 2000\,\text{W}\,\text{m}^{-2}\,\text{K}^{-1}$): $G_{\text{cond}}^{1\text{–}2}$ increases to 141.4 W/K. Converged $T_1 = 678.4$ K. The wafer is better clamped to the ESC.

(c) **Both changes simultaneously:** $T_1 = 675.2$ K. The wafer temperature approaches the ESC setpoint (673 K) with only 2.2 K offset.

(d) **Mode reclassification:** With gold walls, the wafer–wall path shifts from radiation-dominated to borderline ($\Gamma \approx 0.1$); all other paths remain unchanged. The system no longer requires Algorithm 09.A for the wall path but still needs coupled analysis for the wafer–showerhead path ($\Gamma = 1.7$ is unaffected by wall emissivity).



Path	G_c	G_r	Γ	Mode
1-2 (ESC)	35.4	~0	<0.01	Cond
1-3 (Wall)	0	1.09	>>10	Rad
1-4 (Shwr)	**0.19**	**0.32**	**1.7**	**MIXED**
1-5 (Win)	0	0.16	>>10	Rad

Figure 09.11: Expanded thermal network for WE-09.3 (300 mm PECVD chamber). Five nodes: wafer (1), ESC (2), chamber wall (3), showerhead (4), quartz window (5). Solid blue lines: conduction paths with G values. Dashed red lines: radiation paths with G_{rad} values. Purple dashed line: the mixed-mode wafer–showerhead path ($\Gamma = 1.7$). The MPM table (inset) classifies each path. Node 2 (ESC) is the temperature-controlled anchor; Node 1 (wafer) is the unknown solved by Algorithm 09.A.

Conduction Mode (Diffusionics)

Tool selection guide for PECVD/RTP. Table 09.9 reveals common pitfalls in semiconductor thermal modeling:

- **PECVD (300–400 °C):** $\Gamma \approx 0.5$–3 for wafer–chamber paths. Conduction-only models underestimate total conductance by 33–75%. Always use Algorithm 09.A.
- **RTP (900–1100 °C):** $\Gamma \approx 5$–50. At the high end, radiation dominates and a radiation-only model with conduction correction suffices. At the low end (early in the ramp), use Algorithm 09.A.
- **Etch (< 80 °C):** $\Gamma < 0.1$. Conduction-only model is valid. Radiation can be safely ignored.

Key insight: Even in systems where the dominant path (wafer–ESC) is conduction-only, secondary paths (wafer–wall, wafer–showerhead) may be mixed-mode and contribute errors exceeding the uniformity specification. *Always check* Γ *for every path, not just the dominant one.*

09.7 Network Context: Mode-Path Matrix Integration

This chapter has introduced three types of thermal elements that modify the Mode-Path Matrix: radiative coolers (Recipe 09.R1), radiation shields (Recipe 09.R2), and multiphysics cloak subgraphs (Recipe 09.R3). Each element changes G_{rad}, G_{cond}, or both, potentially shifting the mode classification of a path. To ensure that these modifications are documented consistently across chapters, this section defines a standardized *MPM edge specification schema* for mixed-mode paths.

09.7.1 MPM Edge Schema for Mixed-Mode Paths

DR-09.8 — MPM Edge Schema for Mixed-Mode Paths

Every thermal path in the mixed-mode regime ($0.1 < \Gamma < 10$) shall be documented using the following standardized edge specification. This schema ensures that coupled analysis inputs are complete, traceable, and transferable between engineering teams.

Field	**Description**
Edge ID	`Node_i -- Node_j` (e.g., `Wafer -- Wall`)
G_{cond} (W/K)	Conduction conductance (constant or temperature-dependent expression)
$G_{\text{rad}}(T_{\text{ref}})$ (W/K)	Radiation conductance at reference temperature T_{ref}: $G_{\text{rad}} = 4\varepsilon_{\text{eff}} F_{ij} \sigma_{\text{SB}} T_{\text{ref}}^3 A$
ε_{eff}	Effective emissivity for the path (parallel-plate or view-factor weighted)
F_{ij}	View factor
A (m^2)	Effective radiating area
$\Gamma(T_{\text{ref}})$	Mode dominance ratio at T_{ref}: $G_{\text{rad}}/G_{\text{cond}}$
Spectral band	Wavelength range for ε evaluation (e.g., 8–14 μm)
ω_{rec}	Recommended under-relaxation parameter (from Table 09.7)
δG_{rad} (W/K)	Uncertainty in G_{rad} from parameter uncertainties ($\delta\varepsilon$, δF, δT)
Convergence	Algorithm 09.A required? (Yes/No, with expected iteration count)

Example (from WE-09.3, path 1–4):

```
Edge ID:           Wafer -- Showerhead
G_cond:            0.189 W/K
G_rad(673 K):      0.32 W/K
eps_eff:           0.26
F_14:              0.25
A:                 0.0707 m^2
Gamma(673 K):      1.7
Spectral band:     LWIR (8--14 um)
omega_rec:         0.5
delta_G_rad:       +/- 0.05 W/K (from delta_eps = +/- 0.05)
Convergence:       Yes, ~6 iterations
```

09.7.2 Updating the Mode-Path Matrix After Mixed-Mode Analysis

After Algorithm 09.A converges, the Mode-Path Matrix must be updated with the self-consistent values:

1. **Replace initial G_{rad} estimates** with converged values $G_{\text{rad}}(T^*)$. The initial estimates (evaluated at $T^{(0)}$) may differ by 10–30% from the converged values due to the T^3 dependence.

2. **Update** Γ for each path using the converged temperatures. Paths near the regime boundaries ($\Gamma \approx 0.1$ or ≈ 10) may shift classification after convergence.

3. **Identify the critical path.** The path with the highest total thermal resistance

$(1/(G_{\text{cond}} + G_{\text{rad}}))$ is the bottleneck per DR-T.3. In mixed-mode systems, the critical path often differs from the conduction-only prediction because radiation provides a parallel pathway that reduces resistance on some paths but not others.

4. **Document uncertainties** using the δG_{rad} field from the edge schema. These propagate into the critical path identification: if two paths have comparable resistances within their uncertainty bounds, both should be treated as potential bottlenecks.

5. **Iterate per DR-T.6** if modifying the critical path (e.g., adding a radiation shield) could shift the bottleneck to another path.

Multiphysics Integration

The six-step MPM workflow is now complete. With Chapters 5–7 (Part II, $\Gamma < 0.1$), Chapter 8 (Part III, $\Gamma > 10$), and this chapter ($0.1 < \Gamma < 10$), the Mode-Path Matrix framework of Chapter 4 can now handle *any* thermal system, regardless of mode dominance. The six-step workflow (DR-T.1 through DR-T.6) proceeds identically; only the analysis tool changes based on Γ. This is the central promise of the Mode-Path Matrix: a single architectural framework that dispatches to the appropriate physics.

Chapter Summary

This chapter completed the thermal engineering toolkit by addressing the mixed-mode regime ($0.1 < \Gamma < 10$) where Parts II and III must work together:

1. **Daytime radiative cooling** exploits the 8–13 μm atmospheric transparency window to achieve sub-ambient temperatures. The net cooling power balance (Eq. (09.2)) integrates emitted radiation, atmospheric absorption, solar heating, and convective parasitic losses. The convection hard gate (DR-09.7) must be checked before any spectral optimization.

2. **Material systems** span polymer films ($\varepsilon_{\text{eff}} \approx 0.85$, scalable), photonic multilayers ($\varepsilon_{\text{eff}} > 0.92$, precise spectral control), and chalcogenide metasurfaces ($\varepsilon_{\text{eff}} > 0.95$, near-ideal). The quantitative FOM (Table 09.5) enables engineering-optimal material selection balancing performance, durability, and cost.

3. **Radiation shielding** reduces G_{rad} by inserting low-emissivity surfaces; N shields reduce radiation by factor $\sim (N+1)$. Recipe 09.R2 translates shield parameters into MPM edge updates.

4. **Multiphysics cloaking** requires simultaneous control of conduction and radiation pathways. The instrument-consistent metric $R_{\text{cloak,rad}}$ (Metric 09.M) resolves the spectral band and weighting ambiguity. The sensitivity analysis (Table 09.6) shows that κ-only optimization yields poor R_{multi}; the optimal design trajectory trades modest conduction performance for large radiation improvement.

5. **The self-consistency requirement** (§09.5.1) demonstrates that linear superposition $G_{\text{total}} = G_{\text{cond}} + G_{\text{rad}}(T_0)$ fails in the mixed-mode regime due to the T^3 feedback in G_{rad}.

6. **Algorithm 09.A** (the coupled conduction–radiation solver) provides the formal iterative methodology with dual convergence criteria ($r_T < \epsilon_T$ and $r_E < \epsilon_E$), under-relaxation selection (Table 09.7), Newton–Raphson fallback, and error estimation.

7. **Error analysis** shows that single-mode models introduce 9–91% errors in the mixed-mode band, with the systematic error quantified by $\Delta R/R = \Gamma/(1+\Gamma)$ (Figure 09.9).

8. **The MPM edge schema** (DR-09.8) provides a standardized documentation format for mixed-mode paths, ensuring traceability and team transferability.

Table 09.13: Complete set of radiative cooling, shielding, and multiphysics design rules.

Rule	Section	Description
DR-09.1	§09.2.3	For LWIR radiative cooling, chalcogenide glasses (As_2S_3, As_2Se_3) provide ideal $n \approx 2.4$–2.8 for 8–13 μm metasurface resonators. For cost-sensitive applications, use glass-polymer hybrid.
DR-09.2	§09.3.2	Radiation shield count: $N \geq [(1-\mathcal{A})/\mathcal{A}] \cdot [(1/\varepsilon_1 + 1/\varepsilon_2 - 1)/(2/\varepsilon_s - 1)]$. For $\varepsilon_s = 0.05$ and $\varepsilon = 0.9$, $\mathcal{A} < 0.01$ requires $N \geq 3$.
DR-09.3	§09.3.3	Spectral shields must satisfy: $\varepsilon < 0.1$ in thermal band, $\tau > 0.9$ at diagnostic wavelength, plus DR-10.7 thermal cycling stability.
DR-09.4	§09.4.5	Multiphysics cloaking compatibility: emissivity mismatch $\Delta\varepsilon < 0.05$ (Planck-weighted per Metric 09.M); coating thermal short-circuit $\kappa_{\text{coat}} \cdot t_{\text{coat}} < 10^{-3}$ W/K; $R_{\text{multi}} > 0.80$ for effective camouflage.
DR-09.5	§09.5.3	Coupled iteration convergence: use Algorithm 09.A with dual criteria ($r_T < \epsilon_T$ and $r_E < \epsilon_E$). Select ω per Table 09.7. Switch to Newton–Raphson if convergence stalls. Flag non-convergence at $N_{\max} = 50$.
DR-09.6	§09.5.5	Mixed-mode threshold: use coupled analysis whenever $0.1 < \Gamma < 10$. Single-mode error exceeds 9% at $\Gamma = 0.1$ and 91% at $\Gamma = 10$.
DR-09.7	§09.1.4	Convection hard gate: sub-ambient radiative cooling requires $h_{\text{conv}} < h_{\text{crit}} = P_{\text{cool}}/\Delta T_{\text{target}} - 4\varepsilon_{\text{eff}}\sigma_{\text{SB}}T_{\text{atm}}^3$. If $h_{\text{conv}} > h_{\text{crit}}$, reduce convection before optimizing spectral profile.
DR-09.8	§09.7.1	MPM edge schema: every mixed-mode path ($0.1 < \Gamma < 10$) shall be documented with: G_{cond}, $G_{\text{rad}}(T_{\text{ref}})$, ε_{eff}, F_{ij}, A, Γ, spectral band, ω_{rec}, δG_{rad}, and convergence requirement.

Looking Forward

With this chapter, the three-part toolkit is complete. Part II (Chapters 5–7) handles $\Gamma < 0.1$; Part III (Chapter 8) handles $\Gamma > 10$; and this chapter handles $0.1 < \Gamma < 10$. The six-step Mode-Path Matrix workflow of Chapter 4 can now be applied to *any* thermal system, regardless of mode dominance.

Chapter 10 extends the spectral engineering framework to thermophotovoltaics (TPV), where the selective emitter principles developed in Chapter 8 combine with back-surface reflectors and near-field enhancements to convert thermal radiation to electrical power. TPV systems span all three Γ regimes: the emitter–PV gap may be radiation-dominated (far-field) or mixed-mode (near-field with conduction through the gap structure), while the PV cell thermal management is typically conduction-dominated. Algorithm 09.A and the MPM edge schema (DR-09.8) will be directly applied to the TPV thermal network analysis.

Research Roadmap

Table 09.14: Research roadmap for Chapter 9 topics.

Topic	**Current State**	**5-Year Horizon**	**TPD Connection**
Daytime radiative cooling	5–15 °C below ambient	>20 °C via convection suppression + spectral optimization	Ch. 8, Ch. 9
Chalcogenide metasurface coolers	Lab-scale demonstration	Roll-to-roll nanoimprint; cost < $5/m^2	DR-09.1, App. B
Multiphysics cloaking	Steady-state, broadband	Transient + spectrally resolved cloaking	Ch. 5, Ch. 9
Mixed-mode FEA automation	Manual iteration or built-in	Automated Γ-aware solver with convergence reporting	Ch. 4, Ch. 9
Active thermal camouflage	Phase-change materials (VO_2)	Electrically tunable emissivity surfaces ($\Delta\varepsilon > 0.5$)	Ch. 6, Ch. 15
Instrument-consistent cloaking metrics	Ad-hoc definitions	Standardized metric (Metric 09.M) adopted in ASTM/IEEE	Ch. 9, Ch. 12

Problems

1. **Radiative cooling power.** A PDMS/Ag radiative cooler has $\varepsilon_{\text{eff}} = 0.85$ in the 8–13 μm window and $\varepsilon = 0.05$ outside. The solar absorptance is $\alpha_{\text{sol}} = 0.08$. (a) Calculate the net cooling power P_{cool} at $T_{\text{surf}} = T_{\text{atm}} = 295$ K under AM1.5 illumination ($I_{\text{sol}} = 1000$ W/m^2) with no wind ($h_{\text{conv}} = 0$). (b) Calculate the equilibrium ΔT below ambient for $h_{\text{conv}} = 5\ \text{W m}^{-2}\,\text{K}^{-1}$. (c) At what h_{conv} does the radiative cooler fail to achieve any cooling ($P_{\text{cool}} = 0$ at $T_{\text{surf}} = T_{\text{atm}}$)? Relate your answer to DR-09.7. (d) Compute Γ and confirm this is a mixed-mode system.

2. **Radiation shield design.** A furnace door (emissivity 0.9) must be shielded from a hot interior wall ($T = 1000$ K, emissivity 0.8). Polished stainless steel shields ($\varepsilon_s = 0.1$) are available. (a) How many shields are needed to reduce radiative heat transfer by 95% ($\mathcal{A} = 0.05$)? (b) Calculate the temperature of each shield at steady state. (c) Draw the MPM network showing the cascade of radiation resistances and document one edge using the DR-09.8 schema.

3. **Multiphysics cloak assessment.** An SiO_2/Si_3N_4 binary cloak (Chapter 5) is used on a Si substrate. The cloak achieves $R_{\text{cloak,cond}} = 0.92$. (a) Estimate ε_{eff} of the cloak surface at $\lambda = 10$ μm using EMT for the SiO_2/Si_3N_4 composite. (b) If the background is bare Si ($\varepsilon_{\text{bg}} = 0.65$), calculate $R_{\text{cloak,rad}}$ using Metric 09.M (Planck-weighted at 350 K, LWIR) and R_{multi}. (c) Is emissivity coating needed for this cloak, and why? (d) Compare with the Ge/As_2Se_3 cloak case of WE-09.2.

4. **Coupled iteration practice.** A two-node system has $G_{\text{cond}} = 10$ W/K and $G_{\text{rad}}(T) = 4 \times 0.7 \times 5.67 \times 10^{-8} \times T^3 \times 0.01$ W/K. Node 1 is heated at $Q = 50$ W; Node 2 is fixed at 300 K. (a) Calculate Γ at $T_1 = 400$ K. (b) Solve for T_1 using the conduction-only model. (c) Perform three iterations of Algorithm 09.A with $\omega = 0.5$. Record r_T and r_E at each iteration. (d) Compare with the exact solution (solve the quartic equation numerically). (e) Compute the superposition error (Eq. (09.41)) and verify it matches the actual error.

5. **PECVD chamber sensitivity.** In WE-09.3: (a) How would the converged wafer temperature change if the chamber wall emissivity were reduced to 0.3 (gold-coated walls)? (b) If the He backside pressure were increased to give $h_{\text{gap}} = 2000\ \text{W m}^{-2}\,\text{K}^{-1}$? (c) Both changes simultaneously? (d) Discuss whether the mode classification changes for any path, and document the updated MPM using the DR-09.8 edge schema.

6. **Spectral shield design.** Design a spectral shield for a pyrometry port operating at $\lambda = 3.9$ μm on an oxidation furnace ($T = 1000\,^{\circ}$C). (a) Specify the required $\varepsilon(\lambda)$ profile using DR-09.3. (b) Select a material from Chapter 8 (Table 08.3) and compute $\varepsilon_{\text{eff}}^{\text{shield}}$ using Eq. (09.27). (c) Calculate the radiation attenuation factor in the thermal band. (d) Verify the transmittance at the pyrometer wavelength. (e) Integrate the shield into the MPM using Recipe 09.R2.

7. **Γ transition map.** For a Si wafer ($\kappa = 148$ W/mK at 300 K, $\varepsilon = 0.65$) separated from a chamber wall by a 5 mm gas gap (N_2, $\kappa = 0.026$ W/mK at 300 K): (a) Compute Γ for the wafer-to-wall path at $T = 300$, 500, 700, 900, and 1100 K, accounting for temperature-dependent κ_{N_2} and T^3 radiation scaling. (b) Plot Γ vs. T and identify the temperatures at which the path transitions between regimes. (c) At what temperature does the path enter the mixed-mode band? (d) Overlay the single-mode error curve (Eq. (09.51)) on the same plot.

8. **Convection hard gate sensitivity.** A radiative cooler with $\varepsilon_{\text{eff}} = 0.93$ (glass-polymer hybrid, Table 09.5) is deployed on a building roof at $T_{\text{atm}} = 308$ K (hot climate).

(a) Calculate $P_{\text{cool}}(T_{\text{atm}})$ and h_{crit} for $\Delta T_{\text{target}} = 8\,^\circ\text{C}$. (b) Typical roof-level wind gives $h_{\text{conv}} = 8\ \text{W}\,\text{m}^{-2}\,\text{K}^{-1}$. Can the target be achieved? (c) Design a polyethylene wind shield enclosure and estimate the reduced h_{conv}. (d) Recalculate ΔT with the wind shield and compute Γ for the system. (e) Using FOM_{cool} (Eq. (09.21)), compare the shielded glass-polymer system with an unshielded As_2S_3 metasurface.

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Book 3 of the Open-Source Engineering Optics Trilogy
Cross-references: Eikonal Bridge Ch. 6 (thin-film optimization); QFI Ch. 8 (noise budget analogy)